What Second-Order Meta-Gradients Buy in TTT-E2E: Probe Memory at a Language-Modelling Price

Valentin Lapchevskiy
Independent researcher
v.lapchevskiy@gmail.com

Abstract

TTT-E2E meta-learns through the gradient steps writing context into its fast weights. Does second-order (SO) differentiation through them pay for costing 2.9× a first-order (FO) step in wall-clock time? On a from-scratch text model (25.6M non-embedding parameters) with attention barriers, a registered equal-GPU-second test finds FO with detached inner gradients and 2.9× the steps learns a better language model (.43 nats lower adapted loss) but not SO's probe memory: its best memory within SO's loss budget stays .138 [.084, .192] nats below SO's (two-sided 80% interval; retrospective cross-fit), no closer than FO at equal steps. Fresh seeds replicate this (.139 [.072, .206]); a warmup–stable–decay schedule does not raise its probe-memory contrast; only partial SO recovers some. SO's own run was slower than the timing pilot, so FO received ≈92% of SO's realised GPU-seconds. No behavioural (passkey) separation was found. In a toy, attention's copy starves the FO write, blocking recovery; a linear inner loss, applied at evaluation, rescues FO given a key address. On text this signature is absent and the mechanism open.

1 Introduction

Test-time training (TTT) stores a context in fast weights by gradient steps on a self-supervised loss [4, 35, 44]. TTT-E2E [36] uses the model's own next-token loss as that inner loss and meta-trains the whole network through the inner steps. Differentiating through them is second order (SO); dropping those terms (first order, FO) is the standard cheaper alternative [12, 21]. Here an SO step costs 2.891 FO steps of wall-clock time: with a fixed GPU budget, should one buy SO steps or 2.9 times as many FO steps?

We registered this question as a test before reading its equal-GPU-second arm, on a from-scratch text model in which attention barriers leave the fast weights as the only path to distant text (§2). Given SO's GPU-seconds, FO buys language modelling, not SO's probe memory (§3, Fig. 1a): with 2.9× the steps it reaches an adapted loss .43 nats below SO's, yet its best probe memory within SO's loss budget stays .138 nats below SO's, no closer to SO than FO at SO's step count. Fresh seeds replicate this, a warmup–stable–decay schedule does not raise FO's probe-memory contrast, and only partial SO recovers part of the memory. At equal tokens, in a post hoc reading, the memory–loss frontier is two-sided (Fig. 1b): FO owns the low-loss end, SO the high-memory end, and an exchange rate prices the move.

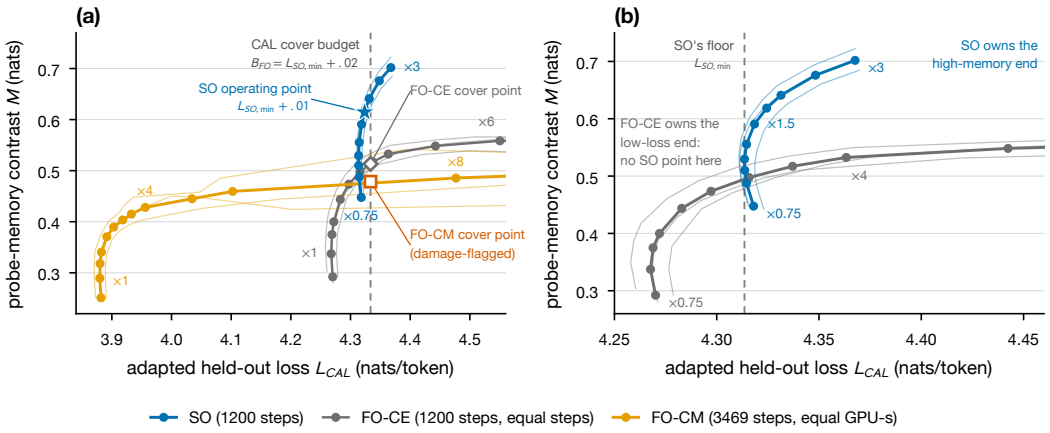


Figure 1: **At equal GPU-seconds FO-CM does not reach SO's probe-memory operating point; at equal tokens the frontier is two-sided.** Original study (mini-m, seeds 0–2; the replication gives the same labels). (a) Equal GPU-seconds: SO@1200, FO-CM@3469 and, grey, FO-CE@1200. (b) Equal tokens (post hoc). Bold: seed means over eval-time η multipliers; thin: seeds. Star: SO's operating point; open square and diamond: FO-CM (damage-flagged) and FO-CE cover points; dashed: budget B_{FO}; R_{cov} is the vertical gap from the star. In-sample selections; cross-fit changes no label (Appendix C).

Why does FO miss the memory? In a controlled key–value toy (§4) the cross-entropy write is proportional to the model's prediction error, so it starves once attention can copy the value. Contrary to our registered prediction this does not start the loss of a binding; it blocks recovery. Once keys have an address, a linear inner loss applied at evaluation rescues FO at FO cost. On text the toy's signature is absent (§5): FO's deficit grows with distance, not with predictability at the write, and the text mechanism remains open.

Contributions.

- A registered, replicated budget test of SO against FO in TTT-E2E on text, read out as a memory–loss frontier with an exchange rate (§3); one FO rule, one from-scratch model size and recipe, probe memory only, and no behavioural separation found.
- A controlled mechanism in which a starved cross-entropy write blocks FO's recovery, with five intervention families and an evaluation-carried first-order fix given a key address (§4); a cause, not the only one, and what starts the loss is not identified.
- A negative transfer result: on text the toy's signature is absent, and the text mechanism is not identified (§5).

2 Setup

TTT-E2E and the first-order rule. TTT-E2E reads a sequence in chunks and stores what it reads in fast weights. Chunk c is predicted with fast weights W_c, which then take one inner step on the next-token cross-entropy (CE) of that chunk:

$$W_{c+1} = W_c - \eta \odot \text{clip}(g_c), \qquad g_c = \nabla_W \ell_{\text{inner}}(\text{chunk } c;\, W_c). \tag{1}$$

Both orders meta-train the slow weights, the initial fast weights W_0 and the step sizes η through the outer loss of every chunk. SO differentiates through every g_c. The FO rule

detaches them, holding the write record fixed, and keeps the direct paths to W_0 and η and the same CE inner loss. It is the only FO rule we trained (Appendix A states what it drops).

Model. All text results use one from-scratch model, *mini-m*: 25.6M non-embedding parameters (91.3M in total), 3.44M of them fast weights in the last two blocks (Table 8). Sequences are 8192 DCLM tokens [19] in chunks of 1024; attention and RoPE reset at chunks 2, 4 and 6, so after a reset only the fast weights reach earlier text. A step is 16 sequences.

Probe memory. A *probe* is the first occurrence, in chunks 4–7, of a 4-gram that also occurs in chunks 0–3, its *sources*. The probe-memory contrast

$$M = L_{\text{fP,focal}} - L_{\text{fP,nonfocal}} \tag{2}$$

is how much more the probe loss rises when the inner-loss terms at a probe's sources are removed (focal) than when a count-matched set of other positions is removed in each chunk (nonfocal). $\text{fP}_{\text{nonfocal}} = L_{\text{fP,nonfocal}}$ is the absolute probe loss with the source written.

Frontier, operating point and cover residual. Every frontier is an eval-time sweep of one fixed checkpoint over inner step-size multipliers η. The adapted loss L (nats/token) is read on 128 held-out documents, split by parity into CAL and TEST; the no-adapt loss H_{LM} on 256; M on 1024 others. With $L_{\text{SO,min}}$ SO's CAL floor over η, and $M_a^*(B)$ the largest M on arm a's piecewise-linear η curve with $L_{\text{CAL}} \le B$, SO's operating point is $L_{\text{SO,min}} + .01$, and an FO arm X is read at the budget $B_{\text{FO}} = L_{\text{SO,min}} + .02$:

$$R_{\text{cov}}(\text{X}) = M^*_{\text{SO}}(L_{\text{SO,min}} + .01) - M^*_{\text{X}}(L_{\text{SO,min}} + .02). \tag{3}$$

With a region of practical equivalence $\text{ROPE}_M = .04$ (registered as $\min(.04, .30\,\Delta M(1))$ from the equal-step default-η gap $\Delta M(1) = .173$; $\lambda^* \cdot \text{ROPE}_M \approx .007$–$.012$ nats at the exchange rate of §3.4), X *does not cover* SO if the lower bound on R_{cov} exceeds .04 (*large* if the mean is at least .08), and *covers* only if the upper bounds on both R_{cov} and Δ_{fn} (X's $\text{fP}_{\text{nonfocal}}$ minus SO's, at the selected points) are at most .04; otherwise the test is *underpowered*. The lift $x = M^*_{\text{FO-CM}}(B_{\text{FO}}) - M^*_{\text{FO-CE}}(B_{\text{FO}})$ compares two FO arms at one budget, so SO cancels. The η that maximises M within the budget is selected using M, so the primary analysis cross-fits: η is selected on one fixed half of the M documents and M reported on the other, then the halves swap. In-sample selection gives the same labels (Appendix C).

Compute match. An SO step costs $r = 2.891$ FO steps of wall-clock time (RTX 4090; each order at its fastest-fitting micro-batch, 2 for SO and 4 for FO; counted FLOPs give 2.02). The equal-GPU-second arm FO-CM therefore trains $N_{\text{CM}} = 1200r = 3469$ steps (455M tokens) against the 1200 (157M) of SO and FO-CE. **Convention:** the budget is SO's GPU-seconds at its fastest-fitting micro-batch (the replication's pilot puts the match at $\rho = .997$). SO itself trained at micro-batch 1, which is slower; against SO's realised training time the 3469-step arms had $\rho \approx .919$ (original study) and $\approx .918$ (replication; Appendix C).

Statistics and status. In three-seed text statistics, bounds are one-sided 90% t bounds over $n = 3$ training seeds ($t = 1.886$), printed as [LB90, UB90]: a *two-sided 80% interval*, so the primary rule is one-sided at $\alpha = .10$; secondary labels are not corrected for multiplicity. Other intervals are labelled where quoted. Results are registered (rule fixed before the data)

Table 1: **Every FO arm misses SO's operating point in both waves.** mini-m, seed means. *Floor*: adapted CAL loss at the η-grid minimum; H_{LM}: no-adapt loss (nats/token). *GPU-s*: GPU-seconds relative to SO@1200 at SO's fastest-fitting micro-batch ($\rho = .997$); against SO's realised micro-batch-1 training the 1.00 arms had $\rho \approx .919$ (original) and .918 (replication). R_{cov} (Eq. (3)) and Δ_{fn} (absolute probe loss with the source written, minus SO's), cross-fit (retrospectively in the original study); [LB90, UB90] is a two-sided 80% interval over three seeds. D: descriptive; other labels registered. x: FO-CM minus FO-CE; x_P: SO-2/16 minus FO-CM. ‡Operating point and ROPE fixed after this arm's frontier had been read (§2).

Arm	Steps (GPU-s)	Floor / H_{LM}	R_{cov} [LB90, UB90]	Δ_{fn} [LB90, UB90]	Label
Original study, seeds 0–2					
SO	1200 (1.00)	4.31 / 4.47	operating point	—	—
FO-CE	1200 (0.35)	4.27 / 4.37	.101 [.076, .127]	.168 [.123, .213]	not covered (large)‡
FO-CM (cosine)	3469 (1.00)	3.88 / 3.94	.138 [.084, .192]	.122 [−.050, .294]	not covered (large)
lift x			−.037 [−.072, −.002]		no closer
Replication, fresh seeds 3–5					
SO	1200 (1.00)	4.31 / 4.46	operating point	—	—
FO-CE	1200 (0.35)	4.26 / 4.36	.061 [.059, .062]	.102 [.085, .120]	not covered (D)
FO-WSD	1200 (0.35)	4.19 / 4.28	.111 [.080, .142]	.168 [.143, .192]	not covered (large; D)
FO-CM (cosine)	3469 (1.00)	3.88 / 3.94	.139 [.072, .206]	.163 [.086, .240]	not covered (large)
FO-WSD	3469 (1.00)	3.91 / 3.96	.213 [.175, .252]	.027 [−.008, .061]	not covered (large)
SO-2/16	2793 (1.00)	3.95 / 4.02	.067 [.022, .112]	.049 [.003, .095]	underpowered
lift x			−.079 [−.146, −.011]		below FO-CE (narrow)
lift x_P			+.072 [+.023, +.121]		higher

unless marked post hoc; our readings are marked "we read". The *original study* (seeds 0–2) was registered before any read of its long FO arm, with in-sample selection ($R_{\mathrm{cov}} = .136$); its cross-fit values are retrospective. Its operating point and ROPE were fixed after the equal-step SO and FO-CE frontiers of seeds 0–2 had been read; only FO-CM was unseen. The *replication* (fresh training seeds 3–5, a denser η grid, two schedule arms and a partial-SO arm) tests the same rule, registered before training, cross-fit included.

3 At equal GPU-seconds, FO buys language modelling, not SO's probe memory

3.1 The answer, and its replication

In the original study FO-CM spent its extra steps on the language model: adapted floor 3.88 nats against SO's 4.31, no-adapt loss 3.94 against 4.47 (Table 1). Probe memory did not follow. FO-CM's best M within B_{FO} stays $R_{\mathrm{cov}} = .138$ [.084, .192] below SO's operating point, so the registered label is *not covered, large*, and FO-CM ends no closer to SO than FO-CE at equal steps ($x = -.037$ [−.072, −.002]). The co-condition $\Delta_{\mathrm{fn}} = +.122$ [−.050, +.294] points the same way but is not separated from zero, so the label rests on R_{cov}.

The replication (seeds 3–5, same data) yields the same label, R_{cov}(FO-CM) = .139 [.072, .206], and there FO-CM's absolute probe-loss gap is separated from zero ($\Delta_{\mathrm{fn}} = +.163$ [+.086, +.240]). All six seed-level residuals of the two waves lie above the ROPE (.071–.190; Fig. 2), and the label holds in every registered sensitivity variant (Appendix C).

Two qualifications bound the claim. It is an operating-point statement: FO-CM's cover points sit at high η, where in the original study its focal loss had risen .10–.47 above its own minimum (the registered damage flag fired on every seed); damage inflates M, so the

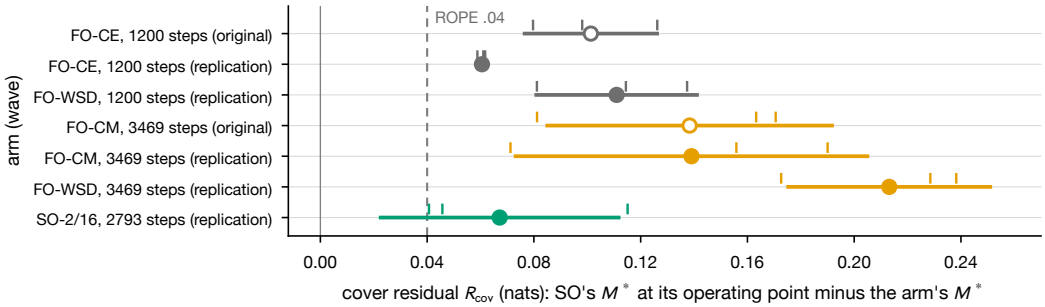


Figure 2: **Every FO arm misses SO's operating point on every seed; only partial SO has a cover-residual interval reaching the ROPE boundary.** R_{cov} per arm and wave (mini-m), cross-fit. Open markers: original study (seeds 0–2); filled: replication (fresh seeds 3–5). Ticks are single seeds; the large marker is the three-seed mean and its bar [LB90, UB90], a two-sided 80% interval. Dashed: $\text{ROPE}_M = .04$. Means, bounds and labels: Table 1.

label is conservative there, but we claim nothing about every η. And with both arms read at the same floor-relative budget $L_{SO,min} + b$, the residual at SO's own floor ($b = 0$) is $\approx .046$, around the ROPE (descriptive): SO's edge grows with the budget (§3.4).

3.2 Not the schedule, not the step count

Is the result an artefact of the cosine schedule stretched to 3469 steps, or of the step count? The replication tested both with a warmup–stable–decay (WSD) schedule at 1200 and 3469 steps. WSD did not raise the probe-memory contrast: at 3469 steps its mean residual is descriptively higher than cosine FO-CM's ($R_{cov} = .213$ [.175, .252]), and at 1200 steps it lies below FO-CE ($\Delta_{shape} = M^*_{\text{WSD-1200}} - M^*_{\text{FO-CE}} = -.050$ [−.080, −.021]; descriptive). In absolute probe loss with the source written, however, WSD-3469 comes within $\Delta_{fn} = .027$ [−.008, .061] of SO: its language model predicts many probes without the source, so part of its lower M is lower headroom (we read this from its probe loss with the source removed, .17–.21 nats below SO's per seed at the selected points). FO-CM's gap persists (§3.1).

More FO steps moved FO's probe-memory contrast further from SO's, not closer. In the replication FO-CM lies below FO-CE by $x = -.079$ [−.146, −.011]. The margin is narrow: UB90 = −.011, seed 4 alone is −.0095, and the original study's x supports only "no closer". Under WSD the step effect on M is clearer: $\Delta_{steps} = M^*_{\text{WSD-3469}} - M^*_{\text{WSD-1200}} = -.102$ [−.114, −.090] (descriptive). The steps went into language modelling: per seed, H_{LM} is 3.93–3.97 for both 3469-step arms against 4.36–4.37 for FO-CE and 4.43–4.50 for SO. We read longer FO training as extending the frontier's low-loss end (§3.4), not its high-memory end.

3.3 What recovers memory

Only one arm moved toward SO. SO-2/16 applies second-order meta-gradients to 2 of the 16 sequences of every batch and first order to the rest, and trains 2793 steps at its own pilot-timed equal-GPU-second horizon. It lies above FO-CM ($x_P = +.072$ [+.023, +.121]), but its cover of SO is underpowered: $R_{cov} = .067$ [.022, .112], and every seed residual (.115, .046, .041) exceeds the ROPE. Descriptively, SO-2/16 recovers $\approx$ 52% of SO's probe-memory edge over FO-CM (.072/.139) while keeping $\approx$ 83% of FO-CM's no-adapt language-model gain over SO, on all three seeds. The two shares combine different measures (no-adapt H_{LM} and the selected adapted M), so this is a trade-off point, not a joint optimum, and only

one row fraction was tested. At equal steps no cheaper lever we tried closes so's default-η memory gap (Appendix C, Table 5), whereas in the toy partial and late so (cf. derivative-order annealing [1]) each restore retention in 12/12 seeds (Appendix B).

3.4 The price: a two-sided frontier and an exchange rate

With tokens rather than GPU-seconds held equal (so and FO-CE, both 1200 steps), the frontier is two-sided (Fig. 1b; a post hoc reading of registered reads). FO-CE owns the low-loss end: its adapted floor is .034–.053 nats below so's per seed and its H_{LM} .096 lower. so owns the high-memory end: at so's floor +.01 its M exceeds FO-CE's at equal L by +.108 (lower bound +.080), and the replication's equal-step residual is $R_{\mathrm{cov}}(\text{FO-CE}) = .061$ [.059, .062]. At so's own floor the edge is $\approx +.02$, so neither order dominates; two post hoc controls argue against inflated η as the source of so's edge (Appendix C).

The exchange rate λ^* is the first $w \in [0, 2]$ at which so's best $L - wM$ equals FO's, over interpolated points with $L \leq L_{\mathrm{SO,min}} + .05$, in nats of per-token loss per nat of per-probe memory. Against equal-step FO-CE, $\lambda^* = .16$–$.30$ in the original study and .24–.44 in the replication; against equal-GPU-second FO-CM, 1.35–1.74 and 1.46–1.77 (per seed; the replication's values are estimates, none censored by the window). λ^* depends on the grid and window and is not portable across probes, sizes or corpora. Cross-segment repeats, the targets only fast weights can serve, are 1.61% of DCLM targets, so a nat of per-probe M worth a nat on each such target gives $w_{\mathrm{tok}} \approx .016$, ten times below every λ^* estimate; we read this as favouring FO's points at both budgets on these curves.

4 Why FO loses memory: a starved write in a controlled toy

The text result says what so buys, not why FO misses it; a toy isolates one reason.

4.1 The echo task and its two barriers

Each echo sequence writes 16 key–value pairs in its first chunk, as events `W v k v`: a write marker, the value, the key and the value again; 300 tokens later a query `Q k` must return v, beyond the 253-token reach of its sliding-window attention, so only a fast-weight head can answer; each sequence draws a fresh map, so no in-weights solution exists. The model is a small transformer with a fast MLP head on top, meta-trained as in §2 for 1500 outer steps; *retention* is query accuracy ≥ 0.90 at step 1500 (chance 1/16; Appendix F).

The task has two barriers. Without a key code, plain FO never binds: so acquired the binding in 7/10 and 7/12 seeds against 0/10 and 0/12 for FO, and key addressability was at chance (1/16) in all 14 probed models that ended unbound, an address barrier rather than a starved write. A frozen random key code added to the fast head's input (+*emb*) removes it and exposes the second barrier: at fixed inner step sizes FO-CE acquired the binding in 16 runs and retained it in none, while so acquired 15 and retained 15/15; a registered fresh-seed replication at $\eta = 40$ gave FO-CE 0/15 against so 12/12 (Appendix B, Table 3).

4.2 One write, followed through

Take the event `W 7 k 7`, which binds key k to value 7. At the key token (a *kv* position, whose next-token target is the written value) the target is the second 7, and attention can copy it from the 7 just before the key. After the chunk is read, the CE inner update associates the

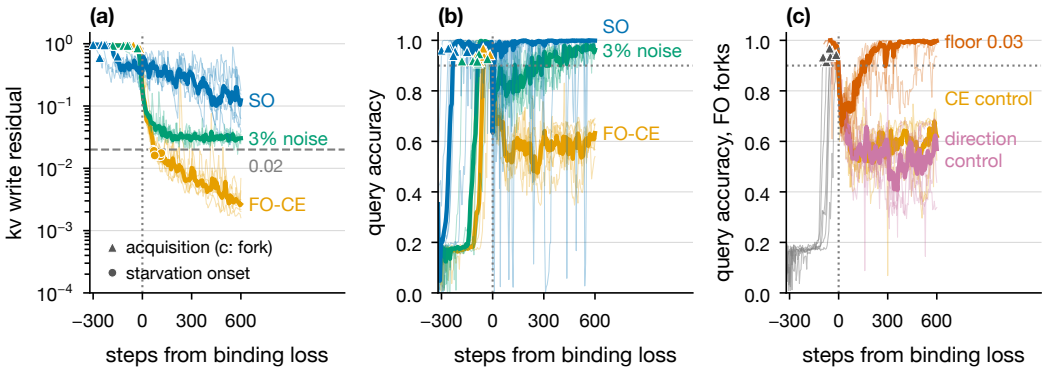


Figure 3: **Every arm loses the binding at first; only FO-CE's kv residual then collapses, and only FO-CE never recovers.** Echo, fixed $\eta = 40$, aligned at each seed's binding loss (dotted). Triangles: acquisition (in (c), the fork); dots: registered starvation onset (two consecutive logs with residual below 0.02), reached only by FO-CE. (a) Median kv write residual (dashed: 0.02). (b) Query accuracy, 6 seeds per arm. (c) Runs forked from an FO seed's main run at acquisition: unmodified CE, kv residual floored at 0.03, same-norm direction control; grey: main run before the fork. Thin: seeds; thick: medians over ≥ 3 seeds. Events registered; alignment and recovery reading post hoc.

representation of k with 7, with a strength proportional to the prediction residual $1 - p(7)$ at that position; the query `Q k`, 300 tokens later, can be answered only from the fast weights. Early in training the copy is poor and the residual gives a useful write; once the copy is reliable, the residual and the write shrink together. Under FO nothing in the outer gradient keeps the write up; under SO the outer loss at the query can. This is an inner-loop analogue of gradient starvation [24]: a pathway that already predicts the target suppresses the gradient another pathway needs. We call it *write starvation*.

4.3 Starvation blocks recovery; it does not start the loss

We registered the prediction that starvation starts the loss, with the residual persistently low at and before the binding is lost. It was *falsified* (0/6 seeds): the residual at the loss was still .14–.63 (.21–.68 in 15 fresh seeds). The dense logs show something else (Fig. 3; alignment post hoc, reading registered and replicated on fresh seeds): every main arm loses the binding at first, and the arms differ in whether it returns. SO recovers within 10–30 steps (11/11) and an arm with 3% copy noise in 12/12, while FO-CE recovers in 0/15 and its residual collapses by two more decades, to 3×10^{-4}–1.6×10^{-3}.

Starvation, once established, therefore *blocks recovery*. A residual floor switched on after the collapse, in a run forked from the starved state, restores the binding in 12/12 fresh seeds against 1/12 for an unmodified fork ($p = 0.00049$). At that dose (.03) the floor is needed in training (the unmodified fork's weights read with it re-acquire in 1/12) and at evaluation (the recovered models read with plain CE: 0/12 bound); at .10 and above it acts partly through the read. Masking the outer CE at the write positions from acquisition on prevents a sustained loss in 14/15 FO seeds (count-matched control 0/15), but this identifies neither the trigger nor a mediator. We read the starved write as *a* cause of FO's failure to recover, not the only one; what triggers the loss is not identified (Appendix B).

Table 2: **Interventions that restore the write, or remove the copy that starves it, restore FO retention; same-count site controls do not.** Echo, FO-CE+emb, paired seeds; the unmodified anchor retained 0/10 each time; $\bar{z}$: mean logit. †Apple MPS; replicated on CUDA (10/10 against 0/10). Full table with exact tests: Appendix B, Table 4.

Intervention	What changes	Retained	Control (retained)
Outer-loss mask at kv†	no direct outer CE at kv	10/10	adjacent targets: 0/10
Copy-edge attention mask	kv cannot attend to its source	10/10	—
Linear inner loss at kv	inner loss at kv is $\bar{z} - z_y$	10/10	next position: 0/10
Copy noise, $q \in \{3, 5, 10\}\%$	copy wrong w.p. q	18/18	$q = 1\%$: 0/6; filler: 0/6
Separate kv step size	own η at kv	8/10, 10/10	—

4.4 Interventions that restore the write restore retention

Five intervention families that restore the write, or remove the copy that starves it, restore FO retention; same-count site controls, where tested, do not (Table 2). This supports the prediction-coupled write as a cause, not the only one: every intervention changes more than the residual, and the registered outer-mask mediator fell short (6/10; 8/10 required).

4.5 The write target, a first-order fix given an address, and two other domains

At matched update norm the *write target* matters: replacing CE by the prediction-independent linear loss (mean logit minus target logit) at every position closes the echo deficit (difference-in-differences +.454 [+.294, +.615], a two-sided 90% t interval over six seeds). Given a frozen key address, the linear inner loss removes FO's retention failure at first-order cost (10/10 at kv only, 10/10 everywhere, 12/12 on fresh seeds). The rescue is carried by evaluation: on fresh seeds, CE-trained models read with the linear loss bind in 12/12 and linear-trained models read with CE in 0/12. It has a price. Without the key code it never binds (0/12; an address-keeping auxiliary loss restores 12/12, a filler control 0/12); on re-assignment it mostly keeps the old map (new-map accuracy .30–.48, where SO reaches .92–.93; descriptive); and switched onto CE-trained 125M weights at test time it cost .85–1.4 nats, so it is not a drop-in text fix. On an in-context Markov task the linear loss beats FO-CE by .026 nats and, on the last quarter of positions, SO-CE by .005, so FO does not fail at memory in general. A toy SO step costs about one FO step, so the toy says nothing about compute.

A symbolic world model. In a symbolic world-model stream (SCM; Appendix E), no fixed FO write rule from a first remedy phase kept all three kinds of location change (copying and uniform-value actions, first bindings at unseen locations), while SO kept every scored type in 6/6 seeds; an FO linear write at the action tokens only, chosen afterwards, then kept all three in 6/6 fresh seeds; we read FO as limited by its write target, not incapable.

A visual task. In a visual hidden-object task (WM2; Appendix E), whose end-to-end arms write with next-frame MSE rather than CE, a registered read on five fresh training seeds confirmed the pattern: first-order MSE fails to keep the hidden location but revises it (5/5), an FO arm scoring only the memory's own contribution keeps it but does not revise (5/5), and SO keeps it and revises only partly (4/5; the fifth revised more than the registered ceiling).

Caveats: not compute-matched (so 2.3–2.9× FO per step); a GRU 12.6× cheaper per step matches so up to 48 frames; the mechanism analyses are post hoc; seeds share one corpus.

5 On text, the toy's signature is absent

The toy predicts that FO's text deficit sits at writes whose value attention could already copy. A registered diagnostic split mini-m's probes by copyability at the write and by source–probe distance; its label is **distance-like**: FO's relative deficit is *smaller* on copyable strata than elsewhere (−.082 [−.092, −.071]), the opposite of the starvation prediction, and grows with distance (+.114). A registered prediction that FO's write-time error on copyable sources falls below .90 of SO's failed (.935; Appendix C). The distance profile is consistent with interference, which we do not claim as its cause.

SO's surplus is not the write direction and only partly the write size. FO's focal write points almost as well toward the later read as SO's (cosine .357 against .370) and is ≈ 19% smaller in norm, but an $\eta \times 1.25$ dose matching SO's 1.21× per-token gain closes only .219 of $\Delta M(1) = .173$, about a fifth (Appendix C).

No behavioural separation was found. The replication's adaptation-attributable passkey advantage for SO over FO-CM is −.21 nats per digit token [−.66, +.24]; its registered validity check passed. An injected key–value binding was not learnable across the barrier under its protocol (so, one seed; Appendix C). A 400-step continuation of the released 125M TTT-E2E checkpoint (one run per order, not compute-matched) opens a small FO deficit, +.0180 [+.0163, +.0197], without the toy's copy signature (contrast +.0069 [−.0109, +.0241]; 90% domain-cluster bootstraps, without training-seed variance; Appendix D).

6 Related work

Test-time training and fast weights. TTT began as self-supervised adaptation to distribution shift [34]. Fast weights are an old idea [3, 31]; linear attention is a fast-weight programmer with a Hebbian or delta-rule write [30, 41, 42], gradient-descent writes have a dual form as attention [13], fast-weight layers can be meta-learned [9], and TTT layers write by gradient steps on a self-supervised loss [4, 5, 11, 35, 44] whose choice is a design axis [6, 38]. Dynamic evaluation [17] takes TTT-E2E's next-token step without meta-training; key–value TTT can act as linear attention [20]. TTT-E2E reports no first-order ablation; it trains 3.4× slower than full attention at 8K, attributed to gradients of gradients [36].

First-order meta-learning. First-order MAML and Reptile avoid second derivatives [12, 21] at a bias [10]; truncated back-propagation [32] makes a different cost–bias trade-off by shortening the differentiation horizon. Derivative-order annealing, first order early and second order later [1], is the precedent for our late and partial SO arms. Implicit gradients [26], SAMA [8] and a corrected first-order estimator [7] were not run. MAML's benefit can be feature reuse [25], short horizons bias meta-optimisation [40], and meta-learners can ignore support data [43]. The closest result, GradMem [18], finds first order collapsing on key–value retrieval; we add a registered, compute-matched text budget test, a controlled mechanism, and a first-order fix given an address.

Competing pathways and classical learning rules. A local next-token inner loss need not favour storing a fact for a distant query; delayed supervision and context distillation supply

that credit [15, 39], and training data decide whether knowledge sits in weights or context [16, 37]. In-context learning can give way to an in-weights solution [33]; our acquire-then-lose curve looks similar, but each echo sequence draws a fresh map and the competitor, a local copy [22], cannot answer the query. The CE write is error-driven like the delta rule, the linear write prediction-independent like a Hebbian one, so write starvation is blocking in an inner loop: a co-present predictor, attention's copy, suppresses learning [14, 29] (an analogue of gradient starvation [24], not an instance of it), and the linear write's weak re-assignment mirrors the Hebbian overwrite problem behind delta-rule fast weights [30]. What is new is what SO's outer gradient adds: it keeps the write up. Recall [2] and binding [27, 28] probes motivate ours; performative prediction [23] frames what FO drops (Appendix A).

7 Limitations and practical reading

The claims are scoped to one FO outer rule (the first-order class is broader), one from-scratch model size and recipe, and probe memory rather than behaviour; a smaller model and the released 125M model are secondary evidence. One recipe served both orders (peak learning rate 3×10^{-3}, cosine): FO-CM stretched only the horizon and warm-ups, no order had its learning rate swept, and WSD tests the schedule's shape, not the peak rate. The attention resets isolate TTT-E2E's beyond-window regime; layouts in which attention competes for the same probes were not tested. Against SO's realised training the long FO arms had $\approx .92$ of its GPU-seconds, and at the counted FLOP ratio FO would get ≈ 2420 steps (1200×2.02, derived); as FO arms at 1200 and 3469 steps both miss SO's operating point, we read the label as unlikely to depend on either (untested). The partial-SO cover is underpowered at $n = 3$, with one row fraction; two schedules were tested; the replication's third training split was rebuilt, with an unverifiable identity to the original's. The toy covers one task family and layout with a frozen key code, and its starved write is a cause, not the only one. A decisive text test would vary write predictability at fixed source–query distance, with matched budgets, fresh seeds and preregistered mediation checks.

Practical reading (we read). Under this model and recipe, measured in an undertrained regime (≈6–18 training tokens per non-embedding parameter), FO is preferable for adapted language-modelling loss; SO reaches the tested probe-memory operating point, which the tested FO budgets did not. Its deployment value is conditional on what the tasks gain from that memory (we found no behavioural separation), and λ^* prices it only on the deployment's own curves; partial SO is a candidate middle point, not a demonstrated one.

8 Conclusion

At equal GPU-seconds on a from-scratch TTT-E2E text model, first-order meta-training buys a better language model, not second-order probe memory, and neither more FO steps nor another schedule raises its probe-memory contrast. In a toy, a CE write starved by attention's copy blocks FO's recovery, and a linear write target applied at evaluation rescues FO given a key address. On text that signature is absent: what second order buys is measured; why is open.

Acknowledgments and disclosure

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Availability

The repository at `https://github.com/gen4sp/second-order-ttt` contains the code and configurations for the text and toy experiments and the result files of every series in Appendix G; per-document read arrays and checkpoints are not included. WM2 and SCM (Appendix E) are separate projects, and only their result files are included. Text data are DCLM-Baseline [19] (`mlfoundations/dclm-baseline-1.0-parquet`; CC BY 4.0, released for research use), tokenised with the Llama 3 tokenizer (Meta Llama 3 Community License). The 125M continuation starts from the checkpoint released with TTT-E2E [36], whose repository states no licence; we therefore do not redistribute that checkpoint or weights derived from it. The PyTorch reference implementation used only in our tests (ttt-ouroboros) is MIT-licensed.

References

[1] A. Antoniou, H. Edwards, and A. Storkey. How to train your MAML. In *ICLR*, 2019. arXiv:1810.09502.

[2] S. Arora, S. Eyuboglu, A. Timalsina, I. Johnson, M. Poli, J. Zou, A. Rudra, and C. Ré. Zoology: Measuring and improving recall in efficient language models. In *ICLR*, 2024. arXiv:2312.04927.

[3] J. Ba, G. Hinton, V. Mnih, J. Z. Leibo, and C. Ionescu. Using fast weights to attend to the recent past. In *NeurIPS*, 2016. arXiv:1610.06258.

[4] A. Behrouz, P. Zhong, and V. Mirrokni. Titans: Learning to memorize at test time. arXiv:2501.00663, 2025.

[5] A. Behrouz, Z. Li, P. Kacham, M. Daliri, et al. Atlas: Learning to optimally memorize the context at test time. arXiv:2505.23735, 2025.

[6] A. Behrouz, M. Razaviyayn, P. Zhong, and V. Mirrokni. It's all connected: A journey through test-time memorization, attentional bias, retention, and online optimization. arXiv:2504.13173, 2025.

[7] E. M. Chayti and M. Jaggi. A new first-order meta-learning algorithm with convergence guarantees. *Transactions on Machine Learning Research*, 2026. arXiv:2409.03682.

[8] S. K. Choe, S. V. Mehta, H. Ahn, W. Neiswanger, P. Xie, E. Strubell, and E. Xing. Making scalable meta learning practical. In *NeurIPS*, 2023. arXiv:2310.05674.

[9] K. Clark, K. Guu, M.-W. Chang, P. Pasupat, G. Hinton, and M. Norouzi. Meta-learning fast weight language models. In *EMNLP*, 2022. arXiv:2212.02475.

[10] A. Fallah, A. Mokhtari, and A. Ozdaglar. On the convergence theory of gradient-based model-agnostic meta-learning algorithms. In *AISTATS*, 2020. arXiv:1908.10400.

[11] G. Feng, S. Luo, K. Hua, G. Zhang, D. He, W. Huang, and T. Cai. In-place test-time training. arXiv:2604.06169, 2026.

[12] C. Finn, P. Abbeel, and S. Levine. Model-agnostic meta-learning for fast adaptation of deep networks. In *ICML*, 2017. arXiv:1703.03400.

[13] K. Irie, R. Csordás, and J. Schmidhuber. The dual form of neural networks revisited: Connecting test time predictions to training patterns via spotlights of attention. In *ICML*, 2022. arXiv:2202.05798.

[14] L. J. Kamin. Predictability, surprise, attention, and conditioning. In B. A. Campbell and R. M. Church, editors, *Punishment and Aversive Behavior*, pages 279–296. Appleton-Century-Crofts, 1969.

[15] J. Kim and T. Kothari. Delayed supervision for test-time language models. arXiv:2609.32312, 2026.

[16] M. Kim, D.-K. Kim, J. Kwon, N. Yang, K. Jung, and M. Cha. How training data shapes the use of parametric and in-context knowledge in language models. arXiv:2510.02370, 2025.

[17] B. Krause, E. Kahembwe, I. Murray, and S. Renals. Dynamic evaluation of neural sequence models. In *ICML*, 2018. arXiv:1709.07432.

[18] Y. Kuratov, M. Kairov, A. Bulatov, I. Rodkin, and M. Burtsev. GradMem: Learning to write context into memory with test-time gradient descent. arXiv:2603.13875, 2026.

[19] J. Li, A. Fang, G. Smyrnis, M. Ivgi, et al. DataComp-LM: In search of the next generation of training sets for language models. In *NeurIPS Datasets and Benchmarks Track*, 2024. arXiv:2406.11794.

[20] J. Liu, S. Elflein, O. Litany, Z. Gojcic, and R. Li. Test-time training with KV binding is secretly linear attention. arXiv:2602.21204, 2026.

[21] A. Nichol, J. Achiam, and J. Schulman. On first-order meta-learning algorithms. arXiv:1803.02999, 2018.

[22] C. Olsson, N. Elhage, N. Nanda, J. Joseph, et al. In-context learning and induction heads. *Transformer Circuits Thread*, 2022. arXiv:2209.11895.

[23] J. C. Perdomo, T. Zrnic, C. Mendler-Dünner, and M. Hardt. Performative prediction. In *ICML*, 2020. arXiv:2002.06673.

[24] M. Pezeshki, S.-O. Kaba, Y. Bengio, A. Courville, D. Precup, and G. Lajoie. Gradient starvation: A learning proclivity in neural networks. In *NeurIPS*, 2021. arXiv:2011.09468.

[25] A. Raghu, M. Raghu, S. Bengio, and O. Vinyals. Rapid learning or feature reuse? Towards understanding the effectiveness of MAML. In *ICLR*, 2020. arXiv:1909.09157.

[26] A. Rajeswaran, C. Finn, S. M. Kakade, and S. Levine. Meta-learning with implicit gradients. In *NeurIPS*, 2019. arXiv:1909.04630.

[27] M. V. S. Ravulapalli and S. Chadha. In-context binding capacity in language models. arXiv:2609.30634, 2026.

[28] M. V. S. Ravulapalli, S. Chadha, and A. Hari. Legible failures: Detecting and repairing in-context binding errors. arXiv:2609.11216, 2026.

[29] R. A. Rescorla and A. R. Wagner. A theory of Pavlovian conditioning: Variations in the effectiveness of reinforcement and nonreinforcement. In A. H. Black and W. F. Prokasy, editors, *Classical Conditioning II: Current Research and Theory*, pages 64–99. Appleton-Century-Crofts, 1972.

[30] I. Schlag, K. Irie, and J. Schmidhuber. Linear transformers are secretly fast weight programmers. In *ICML*, 2021. arXiv:2102.11174.

[31] J. Schmidhuber. Learning to control fast-weight memories: An alternative to dynamic recurrent networks. *Neural Computation*, 4(1):131–139, 1992.

[32] A. Shaban, C.-A. Cheng, N. Hatch, and B. Boots. Truncated back-propagation for bilevel optimization. In *AISTATS*, 2019. arXiv:1810.10667.

[33] A. K. Singh, S. C. Chan, T. Moskovitz, E. Grant, A. Saxe, and F. Hill. The transient nature of emergent in-context learning in transformers. In *NeurIPS*, 2023. arXiv:2311.08360.

[34] Y. Sun, X. Wang, Z. Liu, J. Miller, A. A. Efros, and M. Hardt. Test-time training with self-supervision for generalization under distribution shifts. In *ICML*, 2020. arXiv:1909.13231.

[35] Y. Sun, X. Li, K. Dalal, J. Xu, A. Vikram, G. Zhang, Y. Dubois, X. Chen, X. Wang, S. Koyejo, T. Hashimoto, and C. Guestrin. Learning to (learn at test time): RNNs with expressive hidden states. arXiv:2407.04620, 2024.

[36] A. Tandon, K. Dalal, X. Li, D. Koceja, M. Rød, S. Buchanan, X. Wang, J. Leskovec, S. Koyejo, T. Hashimoto, C. Guestrin, J. McCaleb, et al. End-to-end test-time training for long context. arXiv:2512.23675, 2025.

[37] A. Uzunoglu, B. Van Durme, and D. Khashabi. Information abundance paradox: Long-context training undermines parametric knowledge. arXiv:2608.12218, 2026.

[38] K. A. Wang, J. Shi, and E. B. Fox. Test-time regression: A unifying framework for designing sequence models with associative memory. arXiv:2501.12352, 2025.

[39] Z. Wang, X. Dang, R.-J. Zhu, Z. Wen, H. Fu, W. Chai, and J. D. Lee. Learning what to remember: Test-time training via context distillation. arXiv:2608.01672, 2026.

[40] Y. Wu, M. Ren, R. Liao, and R. Grosse. Understanding short-horizon bias in stochastic meta-optimization. In *ICLR*, 2018. arXiv:1803.02021.

[41] S. Yang, B. Wang, Y. Zhang, Y. Shen, and Y. Kim. Parallelizing linear transformers with the delta rule over sequence length. In *NeurIPS*, 2024. arXiv:2406.06484.

[42] S. Yang, J. Kautz, and A. Hatamizadeh. Gated delta networks: Improving Mamba2 with delta rule. In *ICLR*, 2025. arXiv:2412.06464.

[43] M. Yin, G. Tucker, M. Zhou, S. Levine, and C. Finn. Meta-learning without memorization. In *ICLR*, 2020. arXiv:1912.03820.

[44] T. Zhang, S. Bi, Y. Hong, K. Zhang, F. Luan, S. Yang, et al. Test-time training done right. arXiv:2505.23884, 2025.

A What first order drops, formally

This appendix states what the FO rule drops, and what a minimal model says about the consequence. The statements are derived and checked numerically; the models are model-level accounts, not validated descriptions of the toy's dynamics. This appendix gives the derivations in outline; the numerical checks are released as result files (Appendix G, row "Theory loops A, B").

A.1 First order as a decoupled gradient

The FO/SO distinction has an exact description in the language of performative prediction [23], where deploying a model changes the data it is evaluated on. Here the inner loop's write record plays the role of the induced data.

Definition 1. *Fix outer parameters θ. On a sequence x the inner loop (1) produces the write record $\gamma = \mathcal{G}_\theta(x) = (\mathrm{clip}(g_0), \ldots, \mathrm{clip}(g_{C-1}))$, the clipped inner gradients before multiplication by the step sizes. Let $\ell(x, \gamma; \theta)$ be the outer loss of x with fast weights rebuilt from $W_0(\theta)$, $\eta(\theta)$ and γ, and $D(\theta)$ the law of $(x, \mathcal{G}_\theta(x))$. The decoupled risk and the performative risk are*

$$\mathrm{DR}(\theta_d, \theta) = \mathbb{E}_{(x,\gamma)\sim D(\theta_d)}\, \ell(x, \gamma; \theta), \qquad \mathrm{PR}(\theta) = \mathrm{DR}(\theta, \theta);$$

PR *is the training objective.*

Proposition 1. *(a) For every inner loss and either clip state, the FO meta-gradient equals $\nabla_2\mathrm{DR}(\theta, \theta)$. (b) If the inner loss contains no internal stop-gradient and the inner clip is inactive, the SO meta-gradient equals $\nabla\mathrm{PR}(\theta) = \nabla_1\mathrm{DR}(\theta, \theta) + \nabla_2\mathrm{DR}(\theta, \theta)$.*

Proof sketch. The FO graph detaches each g_c, so its gradient is the partial derivative of ℓ with the write record held fixed, averaged over the current distribution: $\nabla_2\mathrm{DR}$. Without internal stop-gradients and with an inactive clip, SO differentiates $\ell(x, \mathcal{G}_\theta(x); \theta)$ completely, and the chain rule splits the result into the path through γ ($\nabla_1\mathrm{DR}$) and the direct path. With an active detached clip, or a detached reference in the inner loss, SO minus FO is the implemented dropped term rather than the full derivative. □

Both identities agree numerically to 5×10^{-8} on a float64 surrogate, and the replayed losses reproduce the toy's own FO and SO losses to 4×10^{-16}. The correspondence is elementary; it describes precisely what FO drops, not an established causal mechanism. Fixed-point statements hold only for deterministic, unregularised gradient descent with a nonsingular preconditioner, $\theta \leftarrow \theta - \alpha P(\theta) g(\theta)$. There the FO fixed points are exactly the stationary points of the decoupled risk, $\nabla_2\mathrm{DR}(\theta, \theta) = 0$, and, under the conditions of Proposition 1(b), the SO fixed points are exactly the performatively critical points, $\nabla\mathrm{PR}(\theta) = 0$. Neither is a minimisation statement: a stationary point need not minimise $\mathrm{DR}(\theta, \cdot)$ (performative stability), and a critical point need not minimise PR (performative optimality). We do not claim that AdamW endpoints are fixed points of either kind.

A.2 A clean key–pair model

Take V values and K pairs with orthonormal key codes, so that pairs decouple, and $W_0 = 0$. At a write, the logits are $a\, e_v$ (a perfect copy of the value v with gain a), the CE write is $m = \eta(e_v - p)$ with $p = \mathrm{softmax}(a e_v)$, and the query logits are $b\, m$. The outer loss is

$L = \mathrm{CE}(ae_v, v) + \lambda\,\mathrm{CE}(bm, v)$. Write $s = 1 - p_v$ for the write residual and $\beta = b\eta V/(V-1)$; then $s = (V-1)/(e^a + V - 1)$, the query margin is βs and the query CE is $\log(1 + (V-1)e^{-\beta s})$.

Proposition 2 (FO starves at fixed scales)**.** *With η, b, $\lambda > 0$ fixed, the FO field on a is $-s < 0$. Gradient flow gives $\dot{a} = s$ and $\dot{s} = -s^2(1-s)$, so $s \sim 1/t$: the write norm per pair, $\eta s\sqrt{V/(V-1)}$, tends to 0, and the query CE tends to its no-memory value* $\log V$*. Without noise the query accuracy stays 1 at every finite time (the margin βs is positive). With uniform labels, label-independent interference whose argmax is almost surely unique is sufficient for it to tend to chance.*

Proposition 3 (SO escapes above a threshold)**.** *The SO field on a is $s(f(s) - 1)$ with $f(s) = \lambda\beta(1-s)(1-\pi(\beta s))$, $\pi(u) = 1/(1 + (V-1)e^{-u})$, and $f(0) = \lambda b\eta$; f is decreasing on $[0, 1]$. If $\lambda b\eta \le 1$, so starves too. If $\lambda b\eta > 1$, there is a unique residual $s^* \in (0, 1)$ that attracts globally along a. s^* is not monotone in $\lambda b\eta$: with $V = 16$ and $\lambda = 1$ its roots are .00928, .08495, .44969, .41009 and .14706 at $\lambda b\eta = 1.01$, 1.1, 2, 10 and 40.*

Proposition 4 (the starved point is a corner)**.** *Parametrise the copy monotonically by $s \ge 0$ ($s = 0$ a perfect copy) and write the outer loss as $\mathrm{PR}(s) = \ell_W(s) + \lambda Q(M(s))$ with $M(0) = 0$ and one-sided derivatives at 0 up to the order used. If the one-sided slope $\mathrm{PR}'(0^+) = \ell'_W(0) + \lambda Q'(0)M'(0)$ is positive, $s = 0$ is a strict local minimum; if it is negative, $s = 0$ is not a local minimum; if it is zero, higher orders decide. If in addition PR' is continuous on some $[0, s_0)$ (and, at zero slope, PR'' too), a positive slope makes $s = 0$ locally attract so and a negative slope makes it repel so. Under a CE write loss $\ell'_W(0) = 1$, so the starved point is a corner of the decoupled problem, robust to any coupling whose first-order strength is below 1. In the clean model $\mathrm{PR}'(0^+) = 1 - \lambda b\eta$: the corner attracts so if $\lambda b\eta < 1$ and repels it if $\lambda b\eta > 1$; at $\lambda b\eta = 1$ higher orders decide, and there they make it attract, as in Proposition 3.*

The derivations are elementary; their use here is the observation that the CE write slope is exactly 1. What the model does not do matters as much. In the noiseless orthogonal model a shrinking write lowers the margin but leaves argmax accuracy at 1, so a collapse of accuracy needs added interference or read assumptions. Its parameters λ and b are not identified in the toy, and so's small nonzero kv residual in the toy is descriptive, not a test of Proposition 3.

A.3 Code identities and the write factorisation

Two identities constrain how the toy's arms may be read. (i) Under the linear inner loss, everywhere or at the kv positions only, a virtual gain on the kv logits leaves the write record unchanged, so the FO and SO fields agree along it (checked 9/9 with the linear loss at the kv positions only). (ii) In chunk 0, where every echo write sits and the fast weights are still W_0, the adaptation-increment CE (which applies CE to the memory's own logit contribution) has the same logit gradient as the linear loss, so its echo retention shares the linear loss's initial-write mechanism; after chunk 0 the two rules differ. Neither identity says that the linear loss's parameter gradient is independent of learned features.

At a fixed state, take write position i (in chunk 0, fast weights W_0) with target y. Let $o_i \in \mathbb{R}^d$ be the hidden state before the final LayerNorm, $C = I - \mathbf{1}\mathbf{1}^\top/d$ the centring matrix, $\sigma_i = (\|Co_i\|^2/d + \epsilon)^{1/2}$ the stabilised LayerNorm scale (ϵ is LayerNorm's stabiliser), $\hat{o}_i = Co_i/\sigma_i$, Γ the LayerNorm gain and U the untied unembedding. The pull-back of the logits through the final normalisation is

$$J_i = \sigma_i^{-1}\, U\,\mathrm{diag}(\Gamma)\,(I - \hat{o}_i\hat{o}_i^\top/d)\,C.$$

With p_i the predicted distribution, $r_i = 1 - p_{y,i}$ the write residual, $q_i = (p_i - p_{y,i}e_y)/r_i$ the distribution over wrong classes, $\bar{u} = \mathbf{1}/V$ (the logit gradient of the linear loss is $\bar{u} - e_y$) and a_i the fast head's hidden activation, the per-key CE increment of the fast head's output matrix is proportional to $r_i J_i^\top (q_i - e_y) a_i^\top$, and the linear increment, with the same factor, to $J_i^\top (\bar{u} - e_y) a_i^\top$. Hence

$$\frac{\|\text{linear increment}\|}{\|\text{CE increment}\|} = \frac{\kappa_i}{r_i}, \qquad \kappa_i = \frac{\|J_i^\top (\bar{u} - e_y)\|}{\|J_i^\top (q_i - e_y)\|},$$

so at a fixed state the CE write is exactly proportional to r_i and the linear write does not depend on it. κ_i has no universal bound once J_i is applied; the measured κ_i lie in .45–1.41 for FO models and in .62–1.03 for bound SO models. This is a statement about one write at a fixed state, not about the summed write or the training dynamics.

A.4 A reduced model that failed

A second reduced model followed three state variables: copy strength, memory gain and the kv log-step. Under its assumptions starvation is asymptotic and a noise floor allows rebinding, with η and the noise level entering separately. Its registered predictions fared badly, and we report them as negative results. Its homogeneous linear read failed its attribution test (0/9; errors .526–.950 against a .25 tolerance). None of its discriminating fixed-η predictions passed (contrast size, threshold shift and timing). For timing, the model predicted that FO-CE's first binding comes earlier at larger η (225 [200, 275] steps at $\eta = 10$ and 50–150 at $\eta = 40$; 25–75 at $\eta = 160$ in its model table); the observed medians over acquired seeds ordered the other way, 200/325/550 at $\eta = 10/40/160$. Its SO-residual family was withdrawn, since "the SO residual cannot rise" is false even in the model. A 4× kv-class gain lifted FO at $\eta = 10$ to .50 in only 1/6 seeds. And the conjecture that the noise rescue needs a learned rise of η is falsified, because the rescue happens at fixed $\eta = 10$. At the trained endpoints, "later writes are locally costly in FO only" fails too: they are locally costly in SO as well (15/15; the not-positive prediction failed), which removes a candidate signature of what FO cannot see.

B The toy in full

This appendix carries the detail that §4 condenses. Runs are listed in Appendix G, settings in Appendix F.

Unit and events. A *unit* is one seed's main run together with the runs forked from it. Events are read on a 5-step grid. Acquisition is the first log at accuracy ≥ 0.90; the *binding loss* is the first later log that starts two consecutive logs below 0.90; *re-acquisition* is the first log after it that starts two consecutive logs at ≥ 0.90; the *starved state* is the third of the first three consecutive logs after the loss with a median kv residual below 0.01. The registered starvation prediction (residual below 0.02 at the loss, in a run of consecutive logs below 0.02 that began strictly before it) was falsified, 0/6 at every registered threshold and sensitivity. Post hoc, at six FO-CE final checkpoints of the fixed-η runs, the residual predicts the failing keys at chance (AUC .47–.56), and a collapse of the read scale is not established.

Residual floors. The *capped* floor multiplies each kv token's CE term by $\min(1000, \max(1, r_{\min}/r_i))$, so the logit gradient keeps its direction and has at least the size it would

Table 3: **With a frozen key code FO acquires the binding and loses it; SO keeps it.** Echo task, +emb unless noted. Acquired: any quick evaluation ≥ 0.90. Retained: full evaluation at step 1500 ≥ 0.90. The original fixed-η rows pool $\eta \in \{10, 40, 160\}$, six seeds per η; the fresh $\eta = 40$ rows use 16 FO-CE seeds (200–215) and 12 SO seeds (200–211). Counts are runs, and retained counts are over all seeds; the registered fresh FO-CE count is 0/15 among the 15 that acquired.

Inner step size	Arm	Acquired	Retained
learned	FO, no code	0/10	0/10
	SO, no code	7/10	6/10
	FO+emb	10/10	0/10
	SO+emb	10/10	10/10
fixed ($\eta \in \{10, 40, 160\}$)	FO-CE	16/18	0/18
	SO	15/18	15/18
fixed $\eta = 40$, fresh	FO-CE	15/16	0/16
	SO	12/12	12/12

have at residual $r_{\min}$ unless the cap binds. The *cap-free* floor, with residual floor d, adds the gradient of $(d - r_i)_+ \ell_{c,i}$ with the coefficient held constant, where $\ell_{c,i} = \log \sum_{j \neq y} e^{z_j} - z_y$ has logit gradient $q_i - e_y$ (q_i the wrong-class distribution of Appendix A). Since the CE logit gradient is $r_i(q_i - e_y)$, the kv logit gradient becomes $\max(r_i, d)\,(q_i - e_y)$ for every r_i. The *direction control* adds, per sequence, the CE gradient of the 15 chunk-0 positions whose target is the next write marker, scaled to the norm the floor adds. From the starved state, the cap-free floor at $d = 0.03$ (registered) restored re-acquisition in 12/12 units against 1/12 for an unmodified fork ($p = 0.00049$); recovery was training-selective at 0.03, mixed at 0.10 and eval-carried at 0.30, and the recovered models need the floored read (0/12 under plain CE). Prevention from acquisition with a floor at $d = 1$ avoided a sustained loss in 11/12 against 0/12 for the control (write gate passed), but it is partly carried by the floored read (control-trained weights read with the floor: 7/12), so by the registered label it is not kv-specific; at $d = 0.30$ prevention was 12/12 but the write gate failed, so that dose is inconclusive. Masking the outer CE at the write positions from acquisition prevented a sustained loss within 300 steps in 14/15 FO units and 9/12 copy-noise units, against 0/15 and 0/12 for a count-matched control.

Second-order halves. Keeping one path at a time (the error-side variant keeps the error factor's graph and holds the activation factors fixed; the input-side variant the reverse) each retained the binding, 10/10 and 9/10 against 0/10, by different routes. Either path alone suffices; this does not say how SO divides the work.

Copy noise. Replacing the copied value by a wrong one with probability q leaves an irreducible residual; for a uniform wrong value the Bayes residual is $(1 - \lambda^2)\frac{15}{16}$ with $\lambda = 1 - 16q/15$. With a learned step size $q = 1\%$ failed in 6/6 and $q \geq 3\%$ retained in 18/18; at fixed $\eta \in \{10, 40\}$ the threshold lay in $(1\%, 3\%]$. Corrupting the expected number of filler tokens instead of the copy retained 0/6. Within noisy models, keys whose copy source was corrupted were recalled better than clean keys in 24/24 cells.

The load price is fragile. At fixed $\eta = 10$, with $N \in \{16, 32, 64, 128\}$ written pairs (12 fresh seeds), both the linear loss and SO retained 12/12 at 16 pairs, and the linear loss

fell behind at 32 (so 11/12, mean .910; linear 0/12, mean .584). At 64 and 128 neither rule learned the binding. A follow-up did not reproduce the 32-pair price: a same-recipe replicate at step 1500 gave a linear-minus-so mean accuracy of $-.127$ [$-.363$, $+.108$] (two-sided 90% t over 12 seeds; against $-.326$ in the first run), it was ≈ 0 at step 3000 (descriptive), and so at $N = 32$ was bimodal and not reproducible per seed (we attribute this to CUDA nondeterminism). We therefore report the price as fragile.

Address auxiliary and layout bounds. Without the frozen code the linear loss never bound (0/12). An auxiliary outer loss on the key representations kept the address in 12/12, and the plain linear rule then bound in 12/12 (Holm $p = 4.9 \times 10^{-4}$), against 0/12 for the same loss on filler tokens. With CE instead of the linear loss the auxiliary gave the acquire-then-lose shape (acquired 12/12, retained 0/12). so did not learn the binding when writes followed a random prefix (0/6, at settings where it bound plain echo in 3/3), so the causal results hold for one layout: all writes in chunk 0, queries in chunks 5–6, a fast head after the attention stack, a frozen key code, 16 pairs and one budget. Several analyses reuse the same seed families, so pooled counts are not independent replicates; the registered fresh-seed replication reproduced the headline contrasts.

Write-rule control. At matched update norm on echo, the prediction-independent linear loss closed the FO-CE deficit (difference-in-differences $+.454$ [$+.294$, $+.615$], two-sided 90% t over six seeds) with both copy-free control gaps inside the ROPE; an own-residual rule had the same echo effect but failed the control in 4/6, so it is not a clean second instance.

Partial second order in the toy. so on 4 of the 32 sequences of every batch for the full budget retained 11/12 on the screen and 12/12 in the registered wave (23/24); so for steps 600–1500 only (late) retained 12/12 in both waves; so for steps 0–400 only (early, screen) retained 1/12. In this toy an so step costs about as much as an FO step (the partial-so arms took 1.10 and 1.01 of so's time per step), so the toy is not a compute argument.

Table 4: **All toy interventions with their status, mediators and controls.** Retained = query accuracy ≥ 0.90 at step 1500; the anchor (FO-CE+emb) retained 0/10 in each experiment. Causal status: R registered contrast, D descriptive; "supports cause" means the registered contrast came out as predicted. Mediator: the registered mediator check, where one was registered. Eval-carried: on fresh seeds, CE-trained anchors read with the linear loss were bound in 12/12, and linear-trained models read with CE in 0/12. A dash means no value was registered or measured. p: raw one-sided exact McNemar against the paired anchor (the two kv step sizes were registered as one family, Holm $p = 0.0039$). †Apple MPS; replicated on CUDA (10/10 against 0/10).

Intervention	What changes	Ret.	Causal status	Mediator	Eval-carried	p
Outer-loss mask at kv†	direct outer CE at kv removed	10/10	R; mechanism not established	6/10; req. 8/10	—	0.00098
same count, adjacent targets	control	0/10	control	—	—	—
Copy-edge attention mask	attention from kv to source blocked	10/10	R; supports cause	—	—	0.00098
Linear inner loss at kv	inner loss at kv: $\bar{z} - z_y$	10/10	R; supports cause	—	—	0.00098
at the next position	same-count site control	0/10	control	—	—	—
Linear inner loss everywhere	no positional labels	10/10	R; supports cause	—	yes (12/12; 0/12)	0.00098
everywhere except kv	descriptive; never acquires	0/10	D; no acquisition	—	—	—
Copy noise $q \in \{3, 5, 10\}\%$	copied value wrong w.p. q	18/18	R; supports cause	—	—	—
$q = 1\%$	weaker dose	0/6	control dose	—	—	—
filler noise, matched count	control	0/6	control	—	—	—
Separate kv step size	η at kv not shared	8/10, 10/10	R; supports cause	—	—	0.0039, 0.00098
Copy-free task	nothing to copy (descriptive)	10/10	D	—	—	—
Warm-up 600 instead of 200	schedule control	0/10	D; schedule control	—	—	—

C Text protocol, sensitivities and diagnostics

Registrations. The original study's equal-GPU-second test was registered before any read of its 3469-step arm: the decision quantities, labels, η grids, the CAL/TEST split and a language-model forecast. Its operating point ($b = .01$) and ROPE_M were set after a post hoc re-reading of the equal-step SO and FO-CE frontiers of the same seeds (equal-token check, Appendix G), so its FO-CE label is not blind. Its registered analysis selected in-sample; the cross-fit selection of §2 was added afterwards as a retrospective audit (below). The replication was registered before training, with fresh training seeds 3–5 on the same data, packs and evaluation sets; cross-fit selection as primary; denser grids; the wording rules for x ("further" if $\text{UB90}(x) < 0$ on the fresh dense grid, which the replication met; §3.2) and x_P ("higher" if $\text{LB90}(x_P) > 0$); the rule that WSD at 3469 steps covering SO would make the schedule a main finding (it did not cover); a wording gate on the compute match; a passkey estimand; and censoring of λ^* at 2. The registrations were internal documents and were not independently time-stamped.

The forecast. Before any read of the 3469-step arm, a language-model channel was registered: shifting FO-CE's $L(\eta)$ down by at most .15 nats leaves $R_{\text{cov}} \geq .063$ (upper bound $\geq .076$), so within that range the language-model channel alone could not reach a cover, and nothing was extrapolated past the grid. The realised floor gain, $\hat{\delta} = .388$ below FO-CE, lies beyond that grid, so the forecast is stated at its own limit and the result is read on the measured arms. At the default η, M is .29 for FO-CM (.28 at an un-annealed step-2400 checkpoint, descriptive), .34 for FO-CE and .51 for SO; FO-CM's whole $M(\eta)$ curve lies below FO-CE's at every $\eta \leq \times 6$. The schedule-regression check (did FO-CM fail to improve FO-CE's floor or H_{LM}?) did not fire. The in-sample lift $x = -.035$ splits into a language-model channel (+.045: FO-CE's $L(\eta)$ shifted down by FO-CM's floor gain, M unchanged) and an M-curve lift (−.080).

Cross-fit selection. The split is `int(sha1(str(row)).hexdigest()[-1],16) % 2` over the evaluation rows, a hash partition with fold sizes 484 and 540. Selection uses one half (adaptive stopping included) and the report uses the other; the two reports are weighted by report-fold size. In the original study cross-fit moves only seed 2's FO-CM point (one fold selects $\eta \times 6.0$ instead of $\times 5.0$): R_{cov}(FO-CM) becomes .138 [.084, .192] from the registered in-sample .136 [.084, .188], R_{cov}(FO-CE) stays .101 [.076, .127], x becomes −.037 [−.072, −.002] from −.035 [−.067, −.002], and no label changes. For λ^*, $L - wM$ is selected on each training half within $L_{\text{SO,min}} + .05$ and the held-out utility is reported; this is a cross-fitted policy exchange rate, not a newly optimised held-out frontier.

Sensitivity variants. Original study: excluding the 43 near-duplicate M documents (which include the 13 prefix-overlap ones) and three held-out documents gives $R_{\text{cov}} = .132$ (lower bound .081); the refinement worst case (one read at the geometric midpoint of the budget-bracketing interval was not taken; the worst case replaces FO-CM's M^* by the larger bracketing grid M) gives .120 (lower bound .071), with x worst at −.008/+.010/−.058 (mean −.019, upper bound +.019), so only "no closer" is claimed for the original study's x. Operating points $b = .005/.01/.02$ give $R_{\text{cov}} = .113/.136/.166$ (descriptive; not computed for the replication). The same-budget residual $R_B(b)$, SO's M^* minus FO-CM's at the same budget $L_{\text{SO,min}} + b$, is $R_B(0) = .085/.002/.052$ (mean $\approx .046$, around the ROPE)

and $R_B(.01)$ = .171/.083/.156 (mean .137, [LB90, UB90] = [.085, .188]). Replication: in-sample selection gives R_{cov}(FO-CM) = .126 (lower bound .070) and the overlap-drop variant .142 [.079, .205], with all labels unchanged in both. Pooling the six seeds of the two waves (descriptive; the grids differ; one-sided 90% t bounds over six seeds) gives R_{cov}(FO-CM) = .139 (lower bound .109) and $x = -.058$ (upper bound $-.028$). In the replication, Δ_{fn} depends strongly on selection mode (FO-WSD-3469, seed 4: .275 in-sample against .015 cross-fit).

Damage flag, TEST diagnostic and grid tops. The original study registered a damage flag: it fires when the arm's focal loss at its cover point exceeds its own η-minimum by more than .03. For FO-CM it fired on every seed (cover η ×6.45/×7.60/×5.00, rise .38/.47/.10); removing its rise above the $\eta \times 1$ value from $M^*_{\text{FO-CM}}$ gives $R_{\text{cov,mem}} = .45$. Above $\eta \times 8$, M rises only as the model breaks (L_{CAL} 4.8–6.4). The replication did not register a damage flag; its FO-CM cover points sit at η ×5.5–×8.1 (descriptive). At the original study's selected points the TEST loss gap is $-.184$ (SD .142), flagged because for seed 2 the maximum sits inside the budget at ×5 and $L(\eta)$ is steep beyond ×6, so CAL/TEST interpolation is coarse; it is not a gate. In the replication, the adaptive top of FO-WSD-3469 stopped on budget at $\eta \times 12$ for seeds 3 and 5 with margins of +.05 and +.085; had those margins been negative, the open top would have blocked its not-covered label.

η grids. Original study: SO $\{.75, .9, 1, 1.1, 1.25, 1.5, 1.75, 2, 2.5, 3\}$; FO-CE@1200 $\{.75, 1, 1.25, 1.5, 2, 2.5, 3, 3.5, 4, 5, 6, 8\}$; FO-CM the same plus $\{10, 12\}$, with ×8–×12 read unconditionally and the adaptive top applied post hoc by truncation (keep points up to and including the first with $L_{\text{CAL}} \geq B_{\text{FO}}$, or until M falls at two consecutive points; cap ×16). Replication: dense grids with unconditional SO ×3.5/×4 and FO ×.5, and an FO cap of ×12; the adaptive top retains up to ×6, then ×7/8/10/12 through the first $L_{\text{CAL}} \geq B_{\text{FO}}$ or two consecutive falls of M, independently per fit half.

Documents and the third training split. M on 1024 evaluation documents; adapted L on 128 held-out documents, CAL even and TEST odd; H_{LM} on 256 held-out documents. FO-CM's steps 1–1200 reuse FO-CE's rows; only steps 3201–N_{CM} touch the third split C. For the replication, C is a fresh local build with the original extension code; its identity with the original study's C cannot be checked (no hashes of it were kept).

The compute match in detail. The timing pilot (RTX 4090) found SO fastest at micro-batch 2 (6607 ms per step, 14.9 GB) and FO at 4 (2285 ms, 18.8 GB), $r = 2.891$, counted FLOP ratio 2.02, $N_{\text{CM}} = 1200r = 3469$. In the original study the speed-normalised ratio r_{act}/r was 1.0017/1.0034/1.0015 (raw .997/.992/.997), and every seed ran the full N_{CM} steps without a guard trip. The replication's pilot gave $r = 2.901$ and $\rho = 3469/(1200r) = .997$, inside the registered $[.95, 1.05]$ window for the "equal GPU-seconds" wording; per-cell compute gates passed for all long arms (r_{act}/r .985–1.024; replication FO-CM seed 3, at 1.024, is more than 2% from 1, so, as registered, its budget is stated as that actual ratio, which changes no label because the gates block only outside .95–1.05). The SO arm, however, trains at micro-batch 1 ($\approx$ 7.1 s per step against $\approx$ 6.6 at the pilot's setting), so its realised seconds are $\approx 1.09\times$ the budget and the long FO arms received $\approx$ .92 of SO's realised seconds (original study .919, replication $\approx$ .918). "Equal GPU-seconds" holds in the registered sense

only. so-2/16's horizon $N_P = 2793$ was fixed from the same pilot as round$(1200\, t_{\text{so}}/t_P)$, with its so rows at micro-batch 2.

Equal tokens: two post hoc controls. These post hoc controls argue against η-induced damage as the explanation for so's tested high-memory edge. For so (seed 0), $\eta \times 1 \to \times 2$ lowers the absolute probe loss with the source written from 2.246 to 2.093 while the focal loss moves −.014; FO's high-η M is the damage-inflated one ($\eta \times 3 \to \times 4$: focal +.051). A budget-free comparison of absolute probe loss points the same way: at matched L, so's $\text{fP}_{\text{nonfocal}}$ is lower by .109–.195 at $\eta \times 1.5$ (3/3 seeds; at $\eta \times 1$ by +.097/+.010/+.040), and so at $\eta \times 2$ reaches 2.09–2.16 against FO's best on $\eta \times 1$–4 of 2.28–2.34. so's M at its floor +.01 also exceeds FO-CE's maximum M at any $\eta \leq \times 4$ by .056–.085, and at $b = 0$ (so's own floor, at $\eta \times 1$ on the earlier reads) the edge is $\approx +.02$. The floor gap is .037/.052/.050 on the earlier reads of the same checkpoints.

A smaller model. On *mini-s* (5.1M non-embedding parameters), at matched loss FO-CE already matches or exceeds so's probe memory: $\rho_{\text{match}} = -.249$ [−.446, −.052] (registered label: the gap trends up with size). so's high-memory edge there holds only at its $\eta \times 1.5$ grid edge ($b = .005$: lower bound −.021), so it is a grid-edge result. The no-adapt language-model gap between the arms shrinks only from .116 at mini-s to .096 at mini-m. Two sizes are not a scaling series (Fig. 5).

The exchange rate in detail. λ^* is computed on the measured piecewise-linear curves (4001 interpolation points, w swept on 2001 values). In the original study (in-sample), so's optimum against FO-CE is interior ($\eta \times 1.5/\times 1.75/\times 1.75$), so by the registered rule λ^* is not an upper bound there, but it is grid-dependent through interpolation (seed 0 fell .19 → .17 when so's extra points were added); against FO-CM, so's optimum sits at the window edge on all seeds, an upper bound with respect to the window. A lower bound is the floor gap divided by so's largest M on the interpolated feasible set (.713/.693/.683): $\lambda^* \geq .048/.077/.073$ against FO-CE and $\geq .592/.636/.642$ against FO-CM; $w_{\text{tok}} \approx .016$ is below both. Cross-fit values: FO-CE .164/.297/.269, FO-CM 1.352/1.736/1.637. In the replication no value was censored at 2: FO-CE .436/.314/.243, FO-WSD-1200 .593/.500/.487, FO-WSD-3469 1.309/1.226/1.234, FO-CM 1.560/1.770/1.462, so-2/16 1.607/1.990/1.896 (seed 4 just under the limit of 2). At mini-s on its own earlier grid, λ^* was .48/.58/.25 against mini-m's .19/.30/.28, with overlapping seeds.

Candidate ranks (descriptive). The stored reads also rank each probe's target among the distinct probe targets of its document. At $\eta \times 1$ (original study, seed 0), top-1 is .716 with the source written and .666 with it suppressed for so, .689/.661 for FO-CE and .735/.711 for FO-CM: FO-CM ranks targets best in absolute terms, while suppressing the source costs so the most. These ranks are not full-vocabulary accuracy and gate nothing.

The cross-over statistic κ (descriptive). This statistic finds so's surplus over FO-CM keyed to the probe's own context, not to a key-to-value address. With $\Delta\tilde{z}_v(t) = \tilde{z}_v(t \mid \text{nonfocal}) - \tilde{z}_v(t \mid \text{focal})$ and $\tilde{z}_v(t) = z_v(t) - \text{mean}_u z_u(t)$, $\kappa^{\times} = \Sigma S / \Sigma[\Delta\tilde{z}_{y_k}(t_k) + \Delta\tilde{z}_{y_j}(t_j)]$ with $S_{kj} = [\Delta\tilde{z}_{y_k}(t_k) + \Delta\tilde{z}_{y_j}(t_j)] - [\Delta\tilde{z}_{y_k}(t_j) + \Delta\tilde{z}_{y_j}(t_k)]$ over probe pairs (k, j) with $y_k \neq y_j$ in the same document, so any additive candidate effect (frequency, multiplicity, recency) and any position effect cancels. κ_Δ is the same ratio on the cross-over terms of

so minus those of FO. At the default η, κ_Δ for so's surplus over FO-CM is .990/1.004/.992 (lower bound .987); at the selected points so is .96–.97, FO-CM .875–.953 and FO-CE .95–1.00. The registered wording "a context-keyed component" requires a point $\geq .5$ and a lower bound $> .25$; $\kappa \approx 1$ does not separate 4-gram from last-token keying and is not a key-to-value address.

Passkey and the injected binding. The replication's passkey primary estimand is so's adaptation-attributable lift over FO-CM: −.21 nats per digit token [−.66, +.24], reading "interval". Validity at $\eta \times 1$: so lift 3.54 (lower bound 2.93), top-1 .99; FO-CE 4.11 (lower bound 3.70), top-1 1.00. The registered wording is an adaptation-attributable preference for the key's digit tokens behind the barrier, not retrieval. The injected-binding test (mini-m, one seed, so only) was labelled not learnable (its retrieval statistic fell from .004 to .001), with no late storage through 2000 continuation steps; the model learned in-segment verbatim recall through attention and the slow weights. A pilot version of the test failed through a design shortcut (key-agnostic priming). This is not a learnability claim about mini-m (one seed, one protocol, read on test templates; read–write recency interference is a design caveat) and not a starvation test.

What so's surplus is not (detail). FO's focal write points almost as well toward the future read as so's (cosine .357 against .370, $\Delta\cos = +.0137$ [+.0113, +.0161]; registered: the alignment hypothesis is not supported). FO's write is $\approx$ 19% smaller in norm ($\|V\|$ FO/so .81, .80 before the clip). An $\eta \times 1.25$ dose matching so's 1.21× per-token gain closes .219 of $\Delta M(1) = .173$. The linearised shares of the first-order benefit gap (write size .56, read sensitivity .26, direction .10, residual .08) are descriptive, not causal. The output cotangent at the source is .935 [.924, .946] of so's on the copy stratum (90% document bootstrap; .975 on one-source probes), so the registered prediction that it falls below .90 failed, while the weight-space write is .81; we read the write-size gap as a learned gain (the logit-to-fast-weight Jacobian and activations), global rather than focal: so's residual-stream RMS before the final normalisation is about 10% lower everywhere, and the focal-minus-nonfocal factor profile is the same in both arms. The starvation class of a separate registered check of the copy stratum (Appendix G, MINI-E) needed a within-segment read-out quantity that was not logged, so it is not evaluable; with that quantity logged the class could only have been mixed (no-starvation needs a lower bound above .95). One-source probes hold 76% of the probes and 63% of ΔM. On mini-s the copyable-strata contrast is −.022 and the copy-stratum cotangent ratio .947.

Cheap levers at equal steps. Table 5 collects the levers of §3.3. The η dose recovers .219/.365/.619 of $\Delta M(1)$ at $\eta \times 1.25/\times 1.5/\times 2$ (×2 fails the language-model guard, $\Delta L = +.0156$; at matched dose the gap is .197). The σ-CE rule weights each token's inner-loss term by $\min(4, \sigma_t/\sigma_{\text{ref}})$, where σ_t is the RMS of the residual stream before the final norm and σ_{ref} its calibration median; it is adverse (−.187). The inverse weight $\min(4, \sigma_{\text{ref}}/\sigma_t)$ gains .176 at $\Delta L = +.0002$, but its frontier lift is unresolved (.071 [+.063, +.079], fading by $\eta \times 4$). On text the partial-so levers are dose-like; in the toy they are switch-like, but the two measures differ (binary retention against a continuous fraction) and are never put on one axis.

Table 5: **Cheap levers recover part of the equal-step, default-η text gap; none closes it (mini-m).** Each row names its training budget, read-out and denominator: fractions are of the default-η gap $\Delta M(1) = .173$ at step 1200, except late so ($\psi = D/(\text{REF} - \text{FO})$ against a continuation reference at continuation step 300), the inverse-σ lift (ΔM units at matched L) and the residual floors (fraction of each preset's ΔM). LM cost is the adapted-loss rise in nats/token (for partial so, the step time). "Toy" is the toy's version of the lever, measured as binary retention and never pooled with the text numbers. The toy's switch-like rescue does not transfer *under this equal-step protocol*; at equal GPU-seconds, see so-2/16 (§3.3). All rows registered.

Lever	Budget / read-out	Denominator	Recovered	LM cost	Toy
η dose	eval, 1200	of $\Delta M(1)$	.219/.365/.619	.001/.005/.016	—
σ-CE	eval	of $\Delta M(1)$	$-.187$ (adverse)	.001–.003	—
Inverse σ	eval	of $\Delta M(1)$	.176	.0002	—
frontier lift	matched L	ΔM units	unresolved (.071)	—	—
Partial so (2/16 rows)	trained 1200	ϕ, of $\Delta M(1)$	.20	step 1.24× FO-CE	12/12
Late so (300 steps)	continuation 300	ψ	.196	—	late: 12/12
Residual floors	trained and eval	of ΔM	$\leq 3\%$	—	12/12

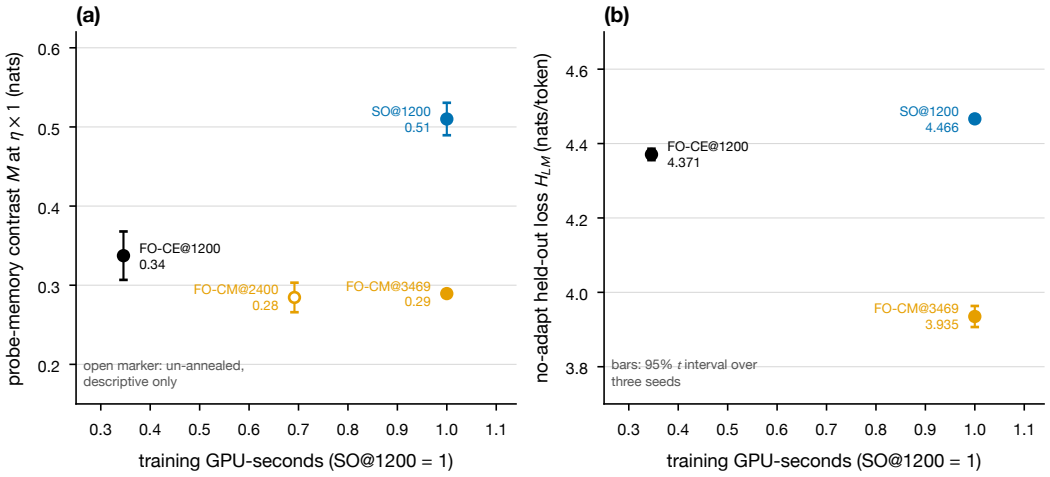


Figure 4: **FO-CM's extra GPU-seconds lowered the no-adapt loss, while its default-η probe memory stayed well below so's (original study, three seeds; descriptive).** (a) M at $\eta \times 1$; (b) no-adapt held-out loss H_{LM}, both against the training GPU-seconds. Filled markers are annealed endpoints; the open marker is the un-annealed FO-CM@2400 checkpoint (descriptive only); no line joins the checkpoints. $r = 2.891$ puts FO-CE@1200 at 0.35, FO-CM@2400 at 0.69 and FO-CM@3469 at 1.0. Bars, two-sided 95% t-intervals over the three seeds.

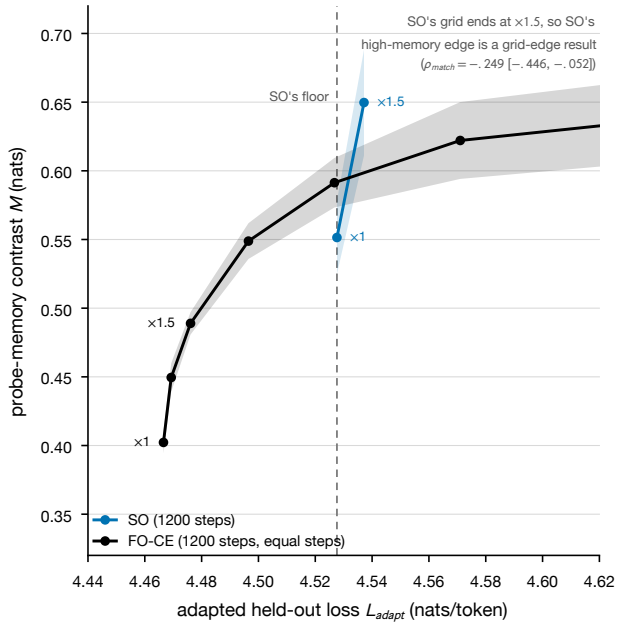


Figure 5: **At mini-s, at matched loss FO-CE matches or exceeds so's probe memory (three seeds; equal tokens).** $\rho_{\text{match}} = -.249$ [−.446, −.052]. so's η grid ends at ×1.5, so so's high-memory edge there is a grid-edge result. Lines join the seed means at each measured η; bands are two-sided 95% t-intervals of M over the three seeds. The x-axis L_{adapt} is the adapted loss on all 128 held-out documents (Fig. 1 uses their CAL half).

D The released TTT-E2E continuation

We continued the released TTT-E2E 125M checkpoint [36] (DCLM, 8K context, step 4799) for 400 outer steps of 64 × 8192 tokens (about 210M tokens) under so and FO-CE, one run per arm, from the same weights and data order. The released peak learning rate 3×10^{-3} destabilised so during re-warm-up (a registered pilot check aborted it at step 30), so both arms used a peak of 10^{-3}. Unlike the toy, the inner clip scale is not detached, so so also differentiates through it. FO opens a small, stable memory deficit, about a third of the registered minimum effect, with no FO language-model advantage. The copy signature the toy predicts does not appear: sources that attention could copy had a write-time inner error about 39% smaller, but equally in both arms. The caveats: one run per arm, a peak learning rate of 10^{-3}, domain-cluster intervals that do not cover training-seed variance, and the comparison is neither compute-matched nor a confirmation of the equal-GPU-second result.

Probes and focal memory. Each 8192-token document is split into two source segments and a probe segment, with the attention context reset at each boundary; a probe is the last token of the first occurrence in the probe segment of a 4-gram that also occurs in a source segment. Focal memory M is the probe loss with the source tokens' inner-loss terms removed, minus the loss with the same number of other source-segment tokens removed in each chunk; the probe segment does not write while M is measured.

Write/read swap. A writer reads the source segments; its fast-weight change from its own W_0 is added to the reader's W_0; the reader then reads the probe with its own slow weights,

Table 6: 125M continuation, endpoint at step 400. $D = L_{\text{FO}} - L_{\text{SO}}$ on barrier probes; $\Delta M = M_{\text{SO}} - M_{\text{FO}}$, the original endpoint estimate; copy contrast $K = D_{\text{copyable}} - D_{\text{barrier}}$. The write/read split comes from a separate writer–reader swap re-evaluation, whose total is .0147 (see text). Intervals are 90% URL-domain cluster bootstraps over documents; with one run per arm they do not cover training-seed variance.

Measure	Value	Interval
D (probe loss gap, FO minus SO)	+.0180	[+.0163, +.0197]
ΔM (focal memory, original endpoint)	+.0143	[+.0134, +.0151]
held-out LM gap (no FO advantage)	+.0007	—
copy contrast K (registered)	+.0069	[−.0109, +.0241]
sensitivity (both sources in segment 2)	+.0186	[+.0011, +.0367] (nominal)
predictable-at-write stratum	−.0142	[−.0173, −.0110]
write-time inner error, FO/SO ratio	.995–1.000	—
swap re-evaluation: write / read	.0070 / .0077	[+.0064, +.0075] / [+.0072, +.0082]
write cosine, FO vs SO	.97–.99	—
focal write size (FO vs SO)	3–8% smaller	—

without writing. Let Q_{ry} be the focal memory with reader r (which also supplies its W_0) and writer y, each SO (s) or FO-CE (f). Writing gets $\frac{1}{2}[(Q_{ss} - Q_{sf}) + (Q_{fs} - Q_{ff})]$ and reading $\frac{1}{2}[(Q_{ss} - Q_{fs}) + (Q_{sf} - Q_{ff})]$, summing to the swap's own $\Delta M = Q_{ss} - Q_{ff}$. On a single GPU this re-evaluation gives a total of .0147 (write .0070, read .0077), against the original endpoint estimate .0143 from the training GPUs; the +.00046 difference is a cross-device re-evaluation difference, within the registered validation tolerance (1.1×10^{-3}). The allocation is convention-dependent: if the writer starts from the reader's W_0, writing gets about 27% and reading 73%.

E Two other domains: SCM and WM2

Both studies are separate projects with their own registrations; neither is compute-matched, and neither is pooled with the echo counts. They bear on the write target, not on the text budget question; keep-versus-revise was not measured on text.

SCM. SCM trains the toy model of §4 on a symbolic world stream whose stored locations change by an agent action that copies its argument (A) or sets a uniform value (Au), and by a first binding at a never-observed location (N). At $\eta = 10$, no fixed FO write rule tried in the first remedy phase keeps A, Au and N together, while SO-CE keeps every scored change type in 6/6 seeds (registered; two rule-based types are descriptive only; at $\eta = 40$ the SO reference was not competent). A first-order linear write applied only at the action tokens, with a per-token weight tied to the token's read-out scale σ_t (capped at 4), was chosen after that phase and then kept A, Au and N in 6/6 fresh seeds (a post-result gate). SCM's registered agency contrast, agent against exogenous changes of matched predictability, was null: in that toy, loss follows predictability at the write, not who caused the change (we read). A residual floor applied at every position from the start of training did not keep first bindings (N, 0 of 6); that is a different task, layout and start, so it is not a site control for §4.3.

WM2: task and arms. WM2 is a visual hidden-object task: a ball is shown, covered by a cup, and then either left in place (*keep*, N) or secretly moved and revealed when a cup is lifted

(*revise*, X). Memory is read beyond a 7-frame attention window, at delays $L = 12/24/48$ frames. Three TTT arms share a fixed step size and write after every frame: E2E-FO, whose inner loss is the whole model's next-frame MSE (WM2's end-to-end target; not the CE of the toy and text), with the write gradient detached; an FO arm whose inner loss scores only the memory's own contribution (AICE-FO); and E2E-SO, with the same MSE inner loss at second order. The registered claims were: E2E-FO fails to keep but revises; AICE-FO keeps but does not revise; E2E-SO keeps ($E_N \geq .90$) and revises only partly ($R_X < .85$).

Table 7: **WM2, registered read on fresh evaluation episodes (original three training seeds).** Every column is an AUROC over the three cup places (chance .5); subscript N: keep episodes, X: revise episodes. The read score α_q projects the predicted frame onto the difference between the twin frames with and without the ball under place q. R: AUROC of α (the behavioural read); E: AUROC of the memory-effect score $\alpha - \alpha(W_0)$, the read minus the same read with the fast weights reset to W_0; X_old: AUROC of the read toward the old location after revision. Thresholds fixed before the data. The read on five fresh training seeds (101–105) confirmed the three claims 5/5, 5/5 and 4/5 against a registered $\geq 4/5$ rule; the SO exception (seed 101, $R_X = .863$) revised more than the ceiling allows.

			AUROC				
Claim	Arm	Seeds	R_N	R_X	E_N	E_X	X_old
fails to keep, revises	E2E-FO	3/3	.50–.56	.87–.92	.54–.60	.93–.95	.31–.34
keeps, does not revise	AICE-FO	3/3	.87–.96	.33–.47	.88–.97	.36–.49	.81–.95
keeps, revises partly	E2E-SO	2/3	.99–1.00	.73/.87/.75	1.00	.74–.86	.63–.77

WM2: caveats. (i) SO costs 2.3–2.9× FO per step and was not compute-matched. (ii) On the training runs' own evaluation episodes (not the fresh episodes of Table 7), a GRU reaches $N = .999$ and $X = .821$, against $X = .767$ for SO, at a step 12.6× cheaper than SO's; at 400 dark frames SO keeps the shown location (1.00 against .59), while the GRU revises better (.90 against .77; the paired difference $-.14$ $[-.24, +.03]$, a 95% bootstrap over seeds and episodes, includes zero). All descriptive. (iii) The fresh-seed claim on SO is partial revision (4/5), and the failing seed failed by revising more. (iv) The mechanism analyses (an intervention on the read and the size of the location signal in the writes) are post hoc; the absence of erasure is not proven; the five fresh seeds share one training corpus. (v) Discriminability of the stored location is not calibrated belief. (vi) A higher-step-size SO arm diverged in all three seeds. (vii) No fixed or switched mix of the two FO targets passed keep, revise and the control in two seeds; such mixes trade keeping for revising along one curve. (viii) Keep-versus-revise was not measured on text. We read WM2's mechanism as only partly the toy's. E2E-FO's total show write is not starved: γ_{show}, the registered ratio of the median update norm of writes targeting the show and cover frames to the median, over episodes, of the largest write between the command and the lift, is .74–.86 over both step sizes. Post hoc, a read-time analysis of the lower-step-size checkpoints finds a weaker location-related signal: the place-mean signal in E2E-FO's show writes is .10–.46 of that in its lift writes, although the location stays decodable from the memory's response (probe .71–.85, chance .33). This is consistent with content-selective suppression of a write that is predictable where it is made, and does not identify a cause during training.

F Hyperparameters and timing

Table 8: Text-model settings, read from each run's configuration. mini-m is every text claim unless stated; mini-s (5.12M non-embedding, 0.88M fast, 37.95M total) appears once in Appendix C; 125M (85.86M non-embedding, 184.36M total) in §5 and Appendix D. Counts from `param_count()`.

Setting	Value
All text	
Model (mini-m)	25.6M non-embedding parameters (91.3M total; the tied 128,256 × 512 embedding is 65.67M), 3.44M fast weights in the last two blocks.
Layout	$T = 8192$, chunk 1024; attention and RoPE reset at chunks 2, 4, 6 ($S = 2048$); global batch 16 sequences (131,072 tokens/step).
Inner loop	CE next-token loss; per-sequence fast weights from W_0; inner step size multiplied by η at eval time; no retraining per frontier point.
Train data	DCLM rows, sequential; FO-CM's steps 1–1200 reuse FO-CE's rows, only 3201–N_{CM} touch the third split.
CAL/TEST	Even/odd indices of 128 adapted-loss documents; η chosen on CAL; M on 1024 separate documents.
Arms	
SO, FO-CE	1200 steps; micro-batch 1 (SO) and 2 (FO-CE).
FO-CM	3469 steps, micro-batch 4; FO-CE's recipe with the horizon and both warm-ups (outer and inner, 347 steps) stretched; cosine; nothing else re-tuned.
FO-WSD	1200 or 3469 steps, micro-batch 4; outer learning rate warm-up over 120 steps to 3×10^{-3}, constant, then linear decay to 10^{-5} over the last 240 steps; inner warm-up .1→1 over 120 steps, then constant, as FO-CE.
SO-2/16	SO on 2 of 16 rows per batch, FO on the rest; $N_P = 2793$ fixed from the pilot before training; warm-up round($.1N_P$), cosine; SO rows at micro-batch 2.
Compute match (RTX 4090)	
Timing pilot	SO fastest at micro-batch 2 (6607 ms, 14.9 GB), FO at 4 (2285 ms, 18.8 GB); $r = 2.891$; counted FLOP ratio 2.02; $N_{\mathrm{CM}} = 1200r = 3469$. Replication pilot: $r = 2.901$, $\rho = .997$.
Realised	Original: $r_{\mathrm{act}}/r = 1.0017/1.0034/1.0015$ (raw .997/.992/.997). Replication: r_{act}/r .985–1.024. Against SO's realised micro-batch-1 training, $\rho \approx .92$.
Tokens	157M (1200 steps), 455M (3469 steps).
125M profile	79.4 s (SO) against 21.6 s (FO) per outer step of 64 × 8192 tokens, one RTX 5090 (3.7×).

Table 9: Toy hyperparameters and run recipe, shared by the echo, copy-free, Markov and load runs; run-level departures are in the last row. s is the run seed.

Setting	Value
Echo task	$T = 512$, $V = 50$ (write and query markers, 16 keys, 16 values and 16 fillers); 16 write events in chunk 0, then a 268-token filler and one query per key, at positions 333–430 in the evaluation set. Each sequence draws its own map from 16 keys to 16 values, uniformly with replacement.
Backbone	Four pre-LN transformer blocks, $d = 128$, four heads, RoPE base 10^4, GELU MLP width 512, causal sliding-window attention of width $w = 64$ and receptive field 253; learned token embedding, final LayerNorm and untied unembedding without bias.
Fast head	$h + W_2\,\mathrm{gelu}(W_1 x + b_1) + b_2$, hidden width 512; $x = \mathrm{LN}(h)$, or $(\mathrm{LN}(h) + \tilde{E}[\mathrm{tok}])/\sqrt{2}$ in +emb arms. $\tilde{E}$ has Gaussian rows scaled to unit RMS, with generator seed $s + 7919$, and is not trained.
Initialisation	Weights $\mathcal{N}(0, 0.02^2)$, biases zero, attention and MLP output projections scaled by $1/\sqrt{8}$, $\rho = 0$, PyTorch seed s. Parameter counts are 938,116 (echo, $V = 50$), 933,508 (Markov, $V = 32$), and 999,556 (load, $V = 290$), plus one parameter for a separate kv step size.
Inner loop	Per-sequence fast weights (W_1, b_1, W_2, b_2) start from learned W_0; $C = T/64$ chunks; mean next-token CE unless the arm changes it; joint gradient norm clipped to 10 per sequence with detached scale. $\eta_i = \eta_{\text{base}} \exp(\rho_i)$, one per fast tensor; learned ρ uses learning rate 10^{-2} and fixed runs use $\eta = \eta_{\text{base}}$.
Outer loop	AdamW with $\beta = (0.9, 0.95)$, peak learning rate 10^{-3}, weight decay 0.01 on matrices, gradient-norm clip 1, batch 32; 200 warm-up steps, then cosine decay from peak at step 200 to 0.1 times peak at the schedule length, which is 3000 for the echo, copy-free and load runs (the Markov exception is in the next row); echo runs stop at step 1500, where the last update uses 0.60 times peak. SO uses `create_graph=True`; FO treats each inner gradient as constant.
Run recipe	Main echo arms use learned η at base 10, fixed-η arms use $\eta \in \{10, 40, 160\}$, and the fresh fixed-η replication uses $\eta = 40$; copy-noise arms use $q \in \{1, 3, 5, 10\}\%$ and filler controls match the expected 5% count; load runs use $N \in \{16, 32, 64, 128\}$ pairs at fixed $\eta = 10$ for 1500 steps; the Markov task uses $T = 1024$, $V = 32$, Dirichlet(0.1) rows, learned η at base 0.3 and 4000 steps, schedule length 4000.
Evaluation	Full echo evaluation uses 256 sequences and 4096 queries; quick evaluation uses 64 sequences; probe evaluation uses 32 sequences and 512 kv events. Training streams are shared across paired arms; copy-noise draws use a separate stream.
Run-level departures	First comparison: 3000 steps. Random-prefix layout: $T = 640$, prefix 64–192, 2000 steps (schedule length 4000) at peak 5×10^{-4}. Step-size sweep: $\eta_{\text{base}} \in \{3, 30\}$, learned, FO only. Linear-loss experiment: one arm with 600 warm-up steps. Load budget extension: 3000 steps, to the end of the schedule.

G Experiment index

Every experiment series of the project, including negative results, with what it was, its outcome, the claim it bears on and the repository folder holding its result files; engineering-only runs (pipeline tests, timing profiles) are omitted. "Original study" is SOQ-N3B; "the replication" is S14-R. S: supported, F: falsified, I: inconclusive, NE: not evaluable.

Series	What it was	Outcome	Bears on	Result files
Text: the budget test and its replication				
SOQ-N3B (+pilot, 3 FO-CM seeds, 4 read jobs)	original study: FO-CM at equal GPU-seconds, frontier reads, κ	not covered (large); x no closer; damage flag on all seeds	§3.1, Table 1	`results/text-original-study/`
A0 cross-fit audit	cross-fit selection on the original study's reads (no training)	labels unchanged; seed 2 FO-CM moves in one fold	§2, App. C	`results/text-crossfit-audit/`
S14-R (12 GPU jobs)	replication: 6 arms × seeds 3–5, training, reads, passkey	FO-CM replicates (.139); x below FO-CE, narrow (§3.2); WSD not covered; SO-2/16 underpowered; passkey interval	§3, §5	`results/text-replication/`
Equal-token check	post hoc re-reading of the original study at equal tokens	two-sided frontier; η-inflation controls	§3.4	`results/text-equal-token-check/`
XPO-0	exposure of cross-segment repeats, DCLM vs code	1.61% of DCLM targets	§3.4	`results/text-repeat-exposure/`
Text: mechanism diagnostics and levers (mini-m, mini-s)				
MINI	from-scratch mini-m/mini-s: SO, FO-CE, residual-floor arms	equal-step gap $\Delta M(1) = .173$; floors $\leq 3\%$	§3.3, Table 5	`results/text-mini-baseline/`
MINI-D	copyable vs distance split of the deficit	distance-like	§5	`results/text-copyability-distance/`
MINI-A	write–read alignment	alignment hypothesis not supported	§5	`results/text-write-read-alignment/`
MINI-E	eval-carried read of the residual floors	$\bar{C}_E \approx 0$; starvation class NE; cotangent .935 (F)	Table 5	`results/text-floors-eval-read/`
MINI-P	partial SO (2/16 rows) at equal steps	$\phi = .20$; no transfer at equal steps	Table 5	`results/text-partial-so-equal-steps/`
MINI-L	300 late SO steps after FO-CE	late-fails ($\psi = .196$)	Table 5	`results/text-late-so/`
TBB	reduced-vocabulary micro testbed	not qualified (relative gap only)	background	`results/text-micro-testbed/`
G0	write-gain split; eval-time η and σ-CE	partial dose; σ-CE adverse	§5, Table 5	`results/text-write-gain/`
G0b	matched-loss frontier; inverse-σ rule	frontier match at SO's default η only; inverse σ .176	§3.4, Table 5	`results/text-matched-loss-frontier/`
G0c	inverse-$\sigma \times \eta$ frontier lift	unresolved (.071)	Table 5	`results/text-inverse-sigma-lift/`
SOQ-FS	mini-s matched-loss frontier	$\rho_{\text{match}} = -.249$; gap trends up with size	App. C	`results/text-mini-s-frontier/`
SOQ-V1	echo-in-text template validity	registered fail, instrument-caused	§5	`results/text-echo-template/`
SOQ-P1-0, -0c	injected-binding pilot	fail through a design shortcut (key-agnostic priming)	App. C	`results/text-injected-binding-pilot/`
SOQ-P1-v3, -v3x, -v3p	injected binding v3; late-storage check; post hoc reads	not learnable; no late storage	§5	`results/text-injected-binding/`
Released 125M model				

Series	What it was	Outcome	Bears on	Result files
P2-0	port and timing profile of the released 125M	SO/FO step time 3.7×	App. F	`results/125m-timing/`
P2-1	400-step continuation, SO vs FO-CE	small FO deficit (D = +.0180); copy contrast not passed	§5, App. D	`results/125m-continuation/`
P2-2	write/read swap	split .0070 / .0077	App. D	`results/125m-write-read-swap/`
Toy: the phenomenon and its barriers				
R1	first FO vs SO comparison (1–2 seeds)	FO does not bind on echo; on the Markov task FO keeps 96% of SO's gain	§4.1	`results/toy-first-comparison/`
R2 bundle	random-prefix layout, Jacobian-SO, copy-free control	anchor gate fired; SO bistable	background	`results/toy-layout-bundle/`
R2b	six-seed contrast	SO binds 4/6, FO 0/6	§4.1	`results/toy-six-seed-contrast/`
R3	random-prefix binding; η sweep	I: no random-prefix arm binds (0/6 each); η sweep 0/3 each	App. B	`results/toy-random-prefix/`
R2-4	write-position ablation (eval only)	kv write necessary and sufficient, 7/7	§4.2	`results/toy-write-ablation/`
R2-4b	key addressability probes	FO: content-failure criterion 4/6, address criterion 0/6 (mixed)	§4.1	`results/toy-key-address/`
R4-0	write-code rank diagnostics	codes collapsed; marker supplies the answer	background	`results/toy-write-code-rank/`
R4-1	key/error/embedding arms ×10	SO 7/10 acq., 6/10 ret.; FO+emb 10/10 acq., 0/10 ret.	Table 3	`results/toy-key-code-arms/`
TH-RC	fixed-η retention	FO-CE 0/18 vs SO 15/18	Table 3	`results/toy-fixed-eta-retention/`
FRS-A	fresh-seed pool (FO 200–215, SO/noise 200–211)	headline replicated	§4.1	`results/toy-fresh-seed-replication/`
Toy: chronology and interventions				
R4-2	outer-loss mask at kv	10/10 vs 0/10; mediator 6/10 (req. 8/10)	Table 2	`results/toy-outer-mask/`
R4-2L	linear inner loss, schedule and copy-free controls	10/10 at kv and everywhere; site control 0/10	Table 2, §4.5	`results/toy-linear-inner-loss/`
R4-3	second-order halves; mask replication	10/10 and 9/10; mask 10/10 vs 0/10	App. B	`results/toy-so-halves/`
R4-4a	copy-edge mask, copy noise, Kamin two-stage [14], kv step size	rows of Table 2	Table 2	`results/toy-copy-interventions/`
R4-4b	AICE rule, Kamin repeat [14]	AICE equals linear in chunk 0; joint criterion not met	App. A	`results/toy-aice-kamin/`
R4-4c	toy rehearsal of the 125M continuation	re-warmed SO 10/10 vs FO-CE 0/10; linear 10/10	App. D	`results/toy-125m-rehearsal/`
BW1a, BW1b	fixed-η copy noise	threshold in $(1\%, 3\%]$	App. B	`results/toy-noise-floor/`
CHR	dense chronology and acquisition forks	starvation-starts prediction F (0/6)	§4.3, Fig. 3	`results/toy-chronology/`
TRG	trigger of the loss	trigger not identified; outer mask prevents 14/15	§4.3	`results/toy-trigger/`
RDO-RPE	recovery dose and prevention floors	floor .03: 12/12 vs 1/12; prevention not kv-specific	§4.3, App. B	`results/toy-recovery-prevention/`
POOL	combined fresh-seed runs (40 main runs, 316 forks)	runs for FRS-A, TRG, RDO-RPE	§4.3	`results/toy-fresh-seed-pool/`
Toy: the write target and its price				
P0.0	Markov-task cost of the linear loss	linear beats FO-CE by .026	§4.5	`results/toy-markov-cost/`
P0.1	mechanism probes (logit split, copy edge, re-assignment)	gate pass 9/9; linear keeps old map	§4.5	`results/toy-mechanism-probes/`
SOQ-D (P3-a)	write-rule control	DiD +.454; own-residual fails control 4/6	§4.5	`results/toy-write-rule-control/`

Series	What it was	Outcome	Bears on	Result files
FRS-B	plain echo × {CE, linear}	linear without code 0/12	§4.5	`results/toy-plain-echo-linear/`
B9	address auxiliary	12/12 vs 0/12 filler	§4.5	`results/toy-address-auxiliary/`
LOD	load scaling, $N = 16$–128	linear behind SO at 32	App. B	`results/toy-load-scaling/`
LOD-X	budget extension to 3000 steps	32-pair price not reproduced (fragile)	App. B	`results/toy-load-extension/`
RTW	per-token write-gain gate; partial-SO screen	gate closed (0/12); $p = 1/8$ 11/12, late 12/12, early 1/12	App. B	`results/toy-partial-so-screen/`
RTW-C	registered partial-SO confirmation	$p = 1/8$ 12/12; late 12/12	App. B	`results/toy-partial-so/`
Other domains and theory				
E1-VE, -D, -D2, -D3, -D4	world-model echo; diagnostics	predecessor of SCM	App. E	`results/visual-echo/`
SCM	symbolic world stream, separate project	SO keeps A/Au/N 6/6; action-selective FO linear write 6/6 (post-result)	§4.5, App. E	`results/scm/`
WM2	visual hidden object, separate project	keep/revise pattern confirmed (5/5, 5/5, 4/5)	§4.5, App. E	`results/wm2/`
Theory loops A, B	performative correspondence; key–pair model; reduced model	derived and checked; reduced model's discriminating predictions F	App. A	`results/theory-checks/`
Parameter check	parameter counts of the text models	counts confirmed	§2	`results/text-param-counts/`